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Sharma

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(54) **SYSTEM AND METHODS FOR TREATING
CANCER CELLS WITH ALTERNATING
POLARITY MAGNETIC FIELDS**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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CPC **A61N 2/004** (2013.01); **A61K 45/06**
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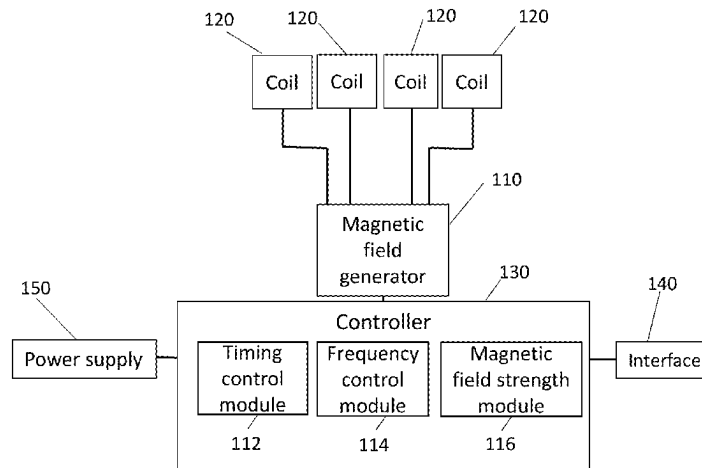
(58) **Field of Classification Search**
CPC A61N 2/00–12; A61N 5/10–1084; A61N
2005/1085–1098

See application file for complete search history.

(57) **ABSTRACT**

Systems and method for destroying or inhibiting cancer cells
and other rapidly-dividing cells include applying AP mag-
netic fields having a defined frequency of 5 Hz-500 KHz and
a field strength of 0.2-5 mT to a portion of the patient's body
that includes the cancer or other rapidly-dividing cells, and
modifying the cancer or tumor microenvironment to
increase the presence of cancer-suppressive cells or decrease
the presence of cancer-promoting cells. In various embod-
iments, the systems and methods may include adjusting the
therapy based on the dosage of a chemotherapy drug, an
immunotherapy drug, a hormone therapy drug, a targeted
therapy drug, or an angiogenesis drug administered to the
patient before, during, or after the application of the AP
magnetic fields.

20 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/828,007, filed on May 30, 2022, now Pat. No. 11,583,692, which is a continuation of application No. 17/396,608, filed on Aug. 6, 2021, now Pat. No. 11,344,740, which is a continuation-in-part of application No. 17/308,019, filed on May 4, 2021, now Pat. No. 11,344,739, which is a continuation of application No. 16/784,239, filed on Feb. 6, 2020, now Pat. No. 11,027,143.

- (60) Provisional application No. 63/127,129, filed on Dec. 17, 2020, provisional application No. 63/063,198, filed on Aug. 7, 2020, provisional application No. 62/802,685, filed on Feb. 7, 2019.

(51) **Int. Cl.**

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A61B 5/055 (2006.01)

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CPC *A61B 5/055* (2013.01); *A61B 6/032* (2013.01); *A61B 6/037* (2013.01); *A61B 6/50* (2013.01); *A61N 5/10* (2013.01)

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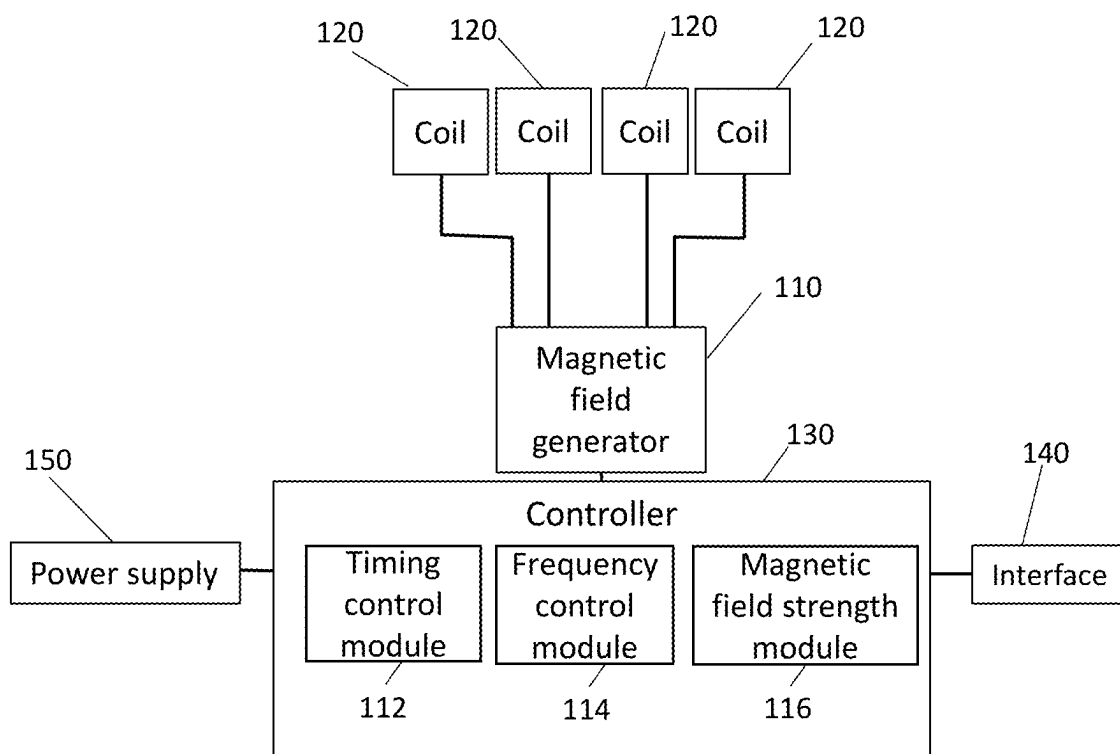


FIGURE 1

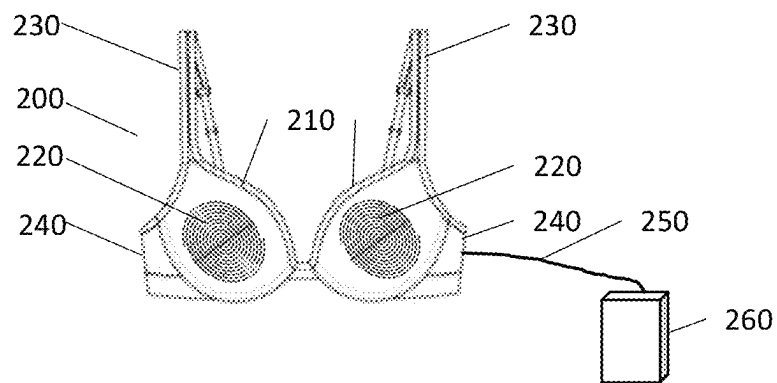


FIGURE 2

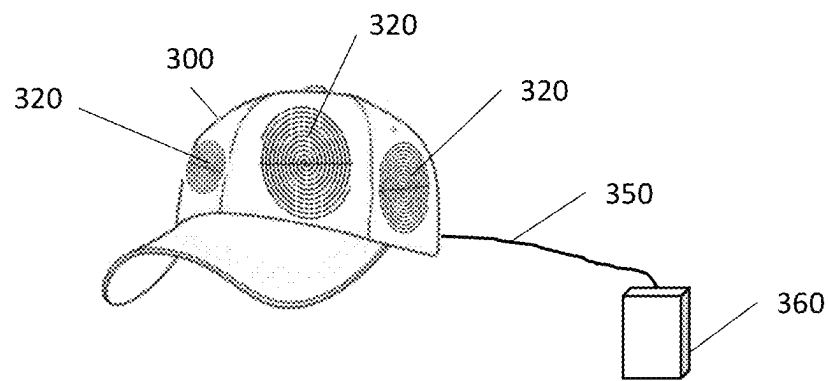


FIGURE 3

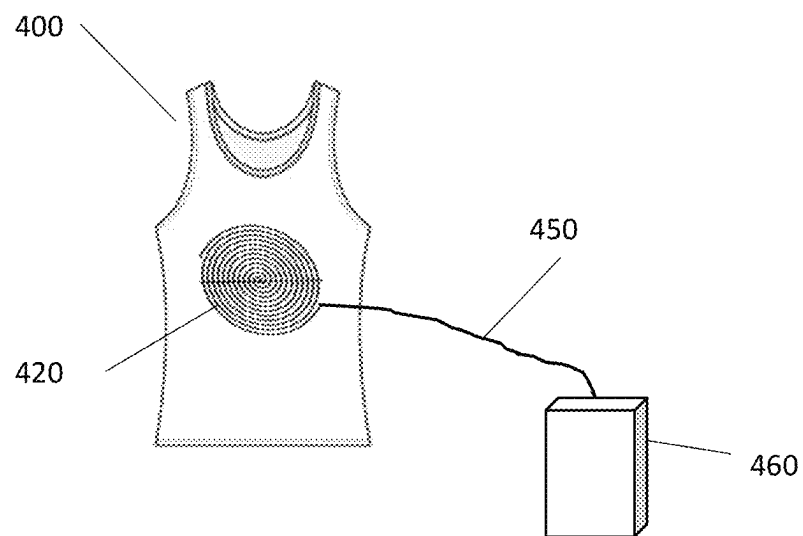


FIGURE 4

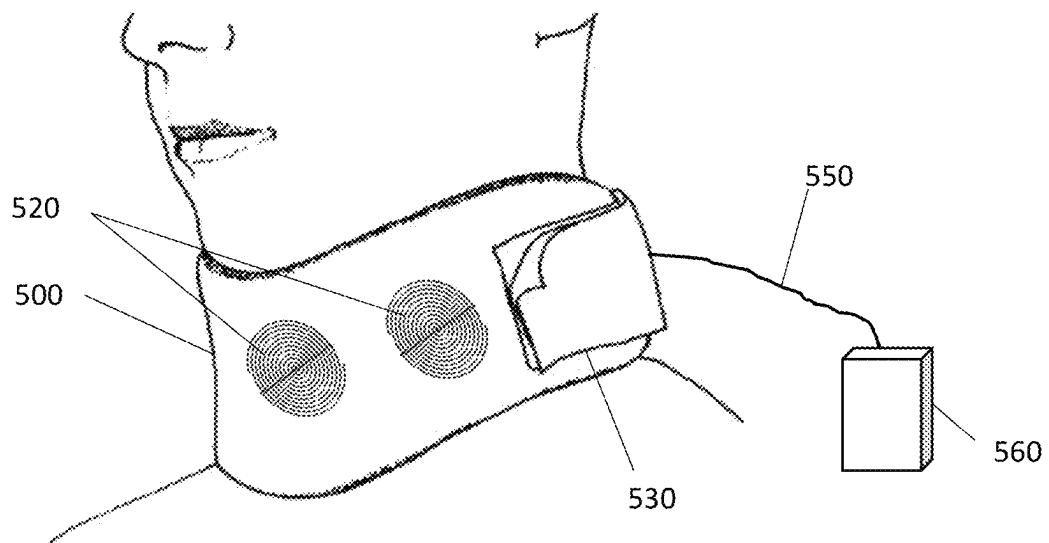


FIGURE 5

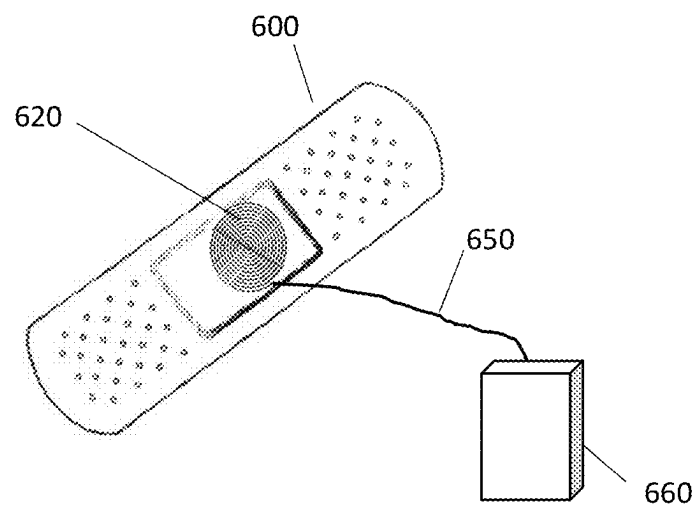
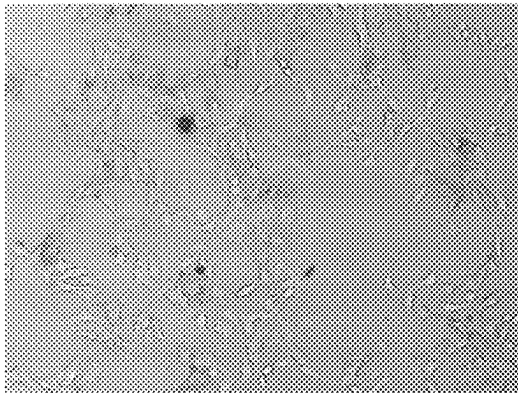
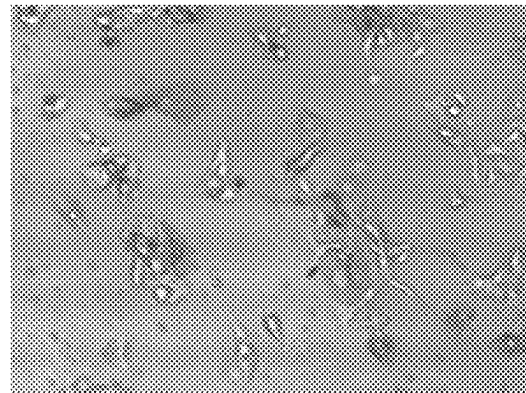


FIGURE 6



Control @10x magnification

FIGURE 7A



Cells exposed to 24 hour magnetic fields @10x magnification showing cell death

FIGURE 7B

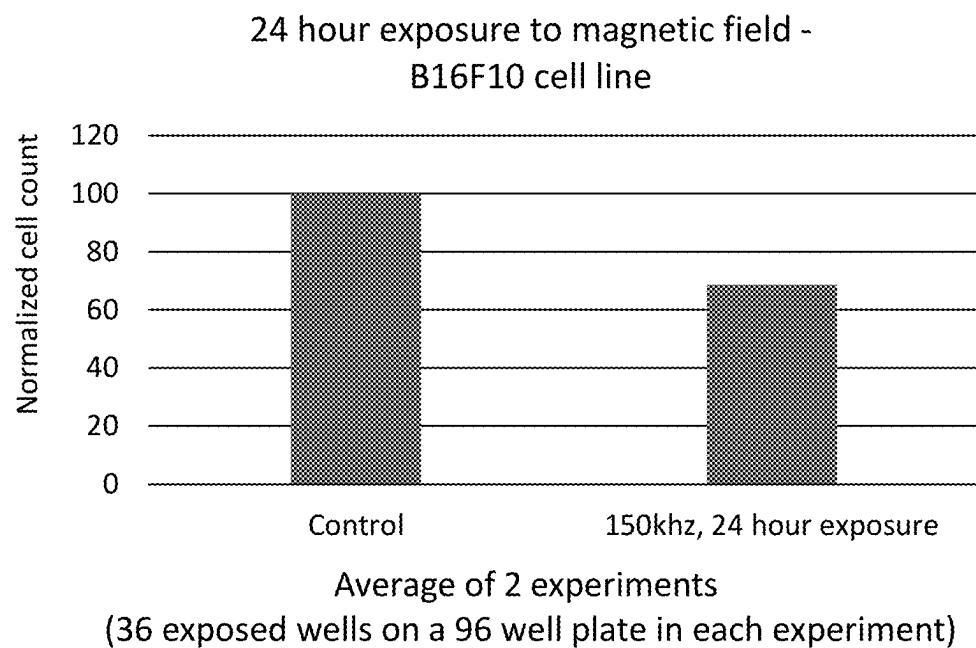


FIGURE 8

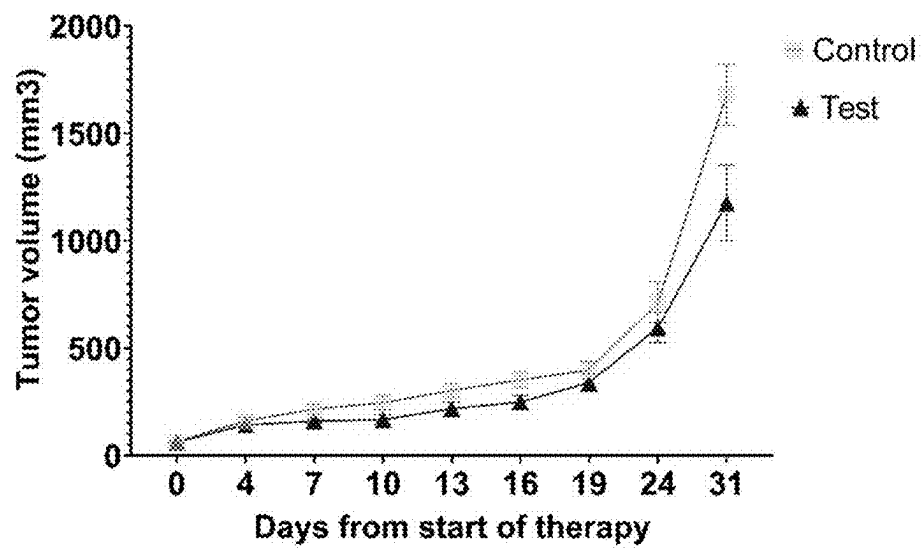


FIGURE 9

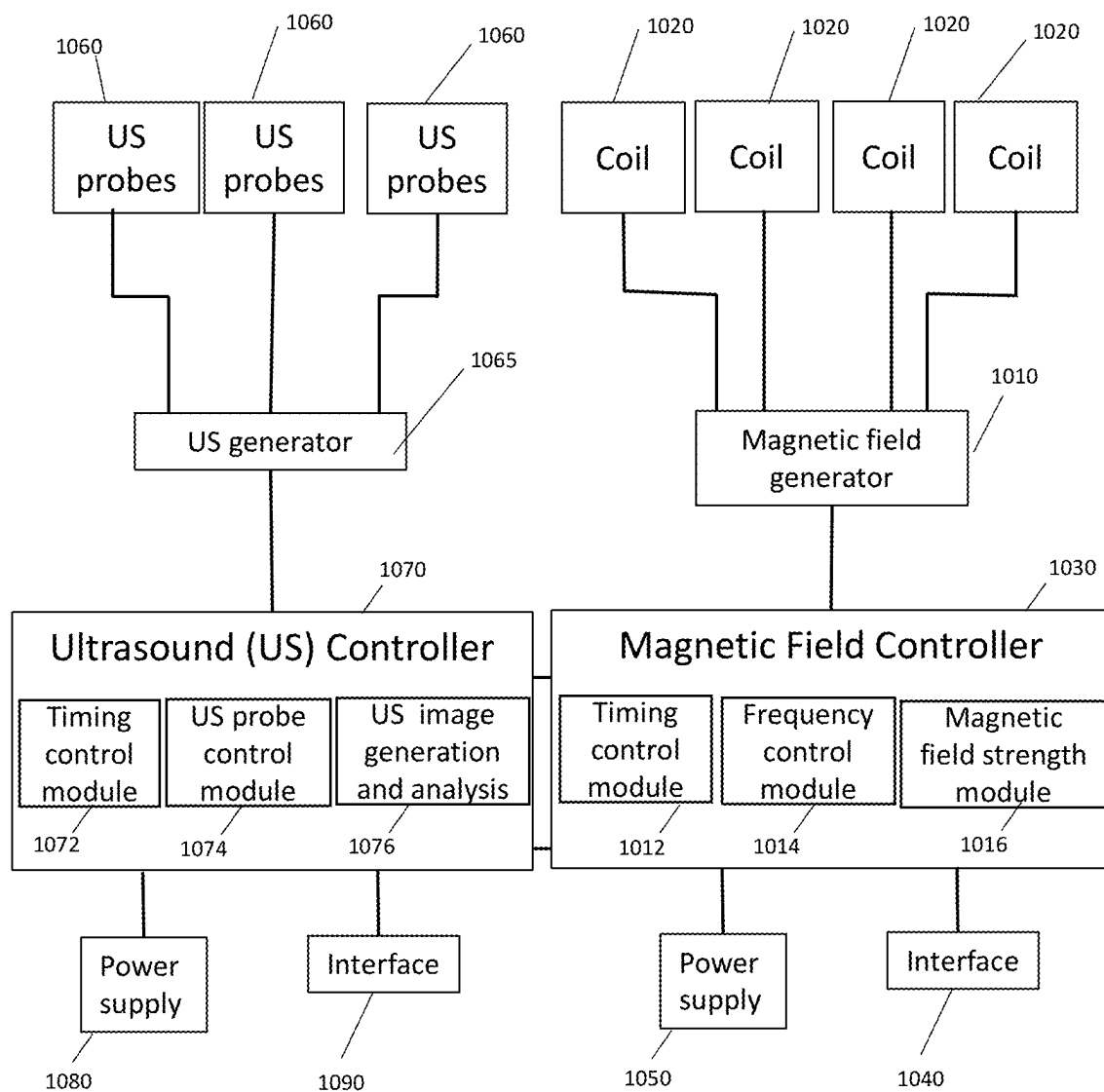


FIGURE 10

Cell line	Exposure frequency	Exposure maximum field strength	Results
MB231 (TNBC cell line)	525 kHz	78uT	Similar cell growth in MTF group and control group
MB231 (TNBC cell line)	745kHz	36uT	Similar cell growth in MTF group and control group
MB231 (TNBC cell line)	20kHz	275uT	Statistically significant reduction in cell growth in MTF compared to the control group (15% reduction, p-value = 0.005).
B16F10 (Melanoma cell line)	150kHz	800uT	Statistically significant reduction in cell growth in MTF compared to control group (31% reduction, p-value < 0.0001)

FIGURE 11

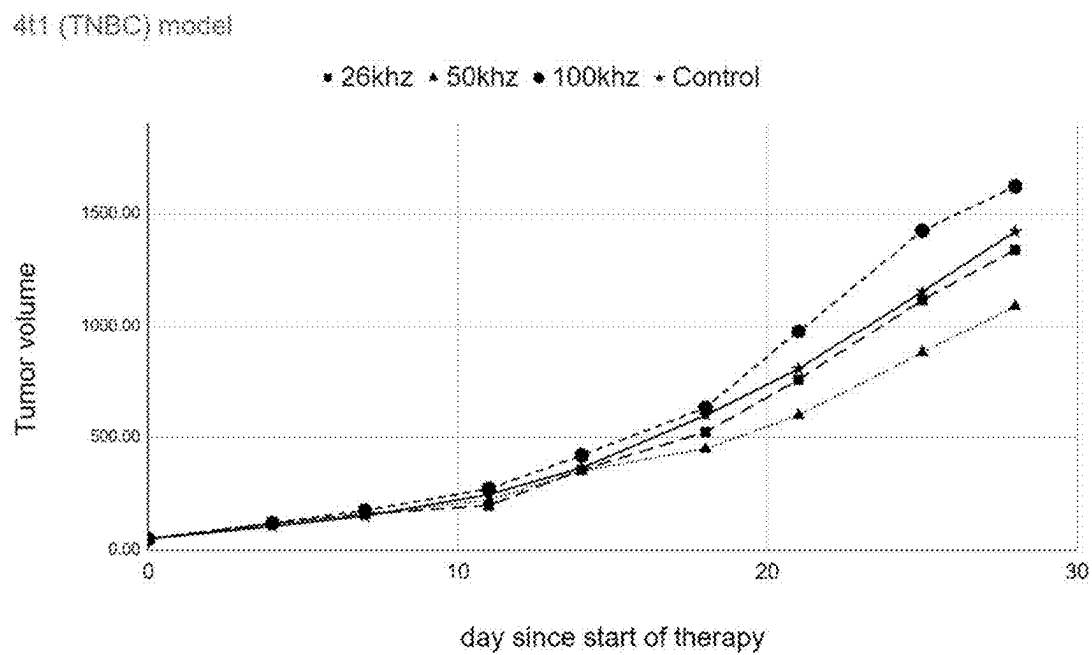


FIGURE 12

Colon cancer model (CT26)

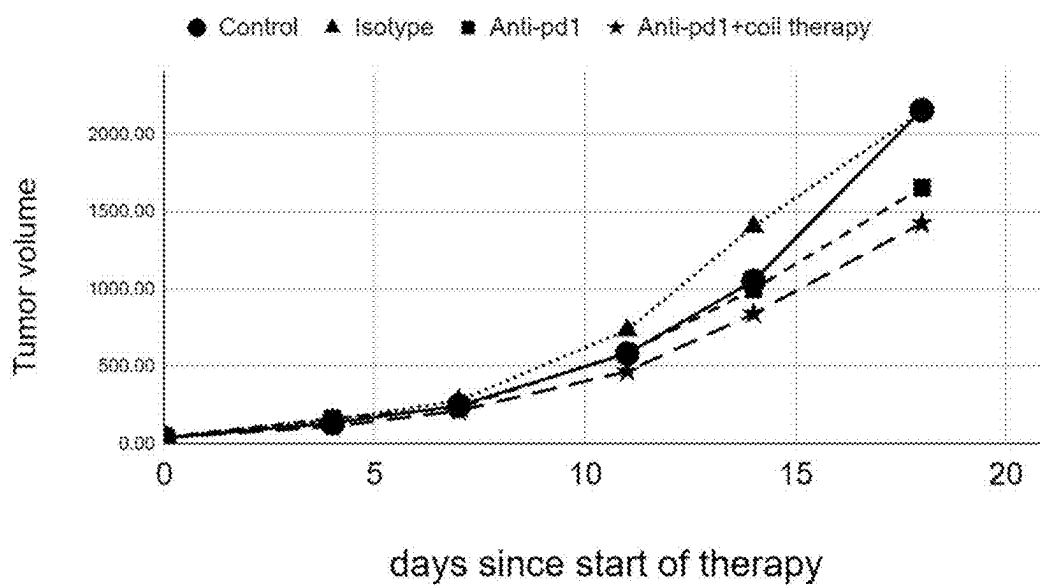
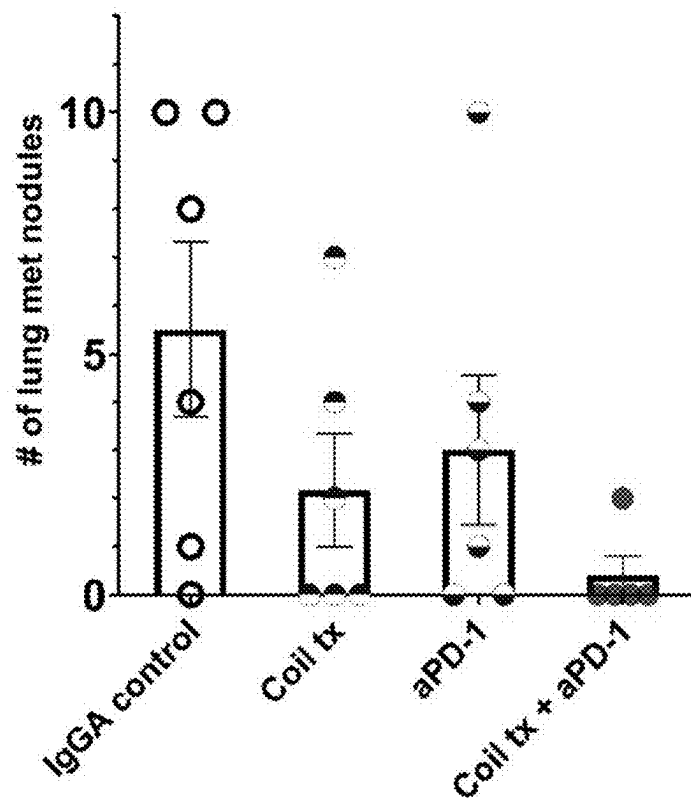


FIGURE 13



both for representation and averaging >10 met nodules were considered as 10

FIGURE 14

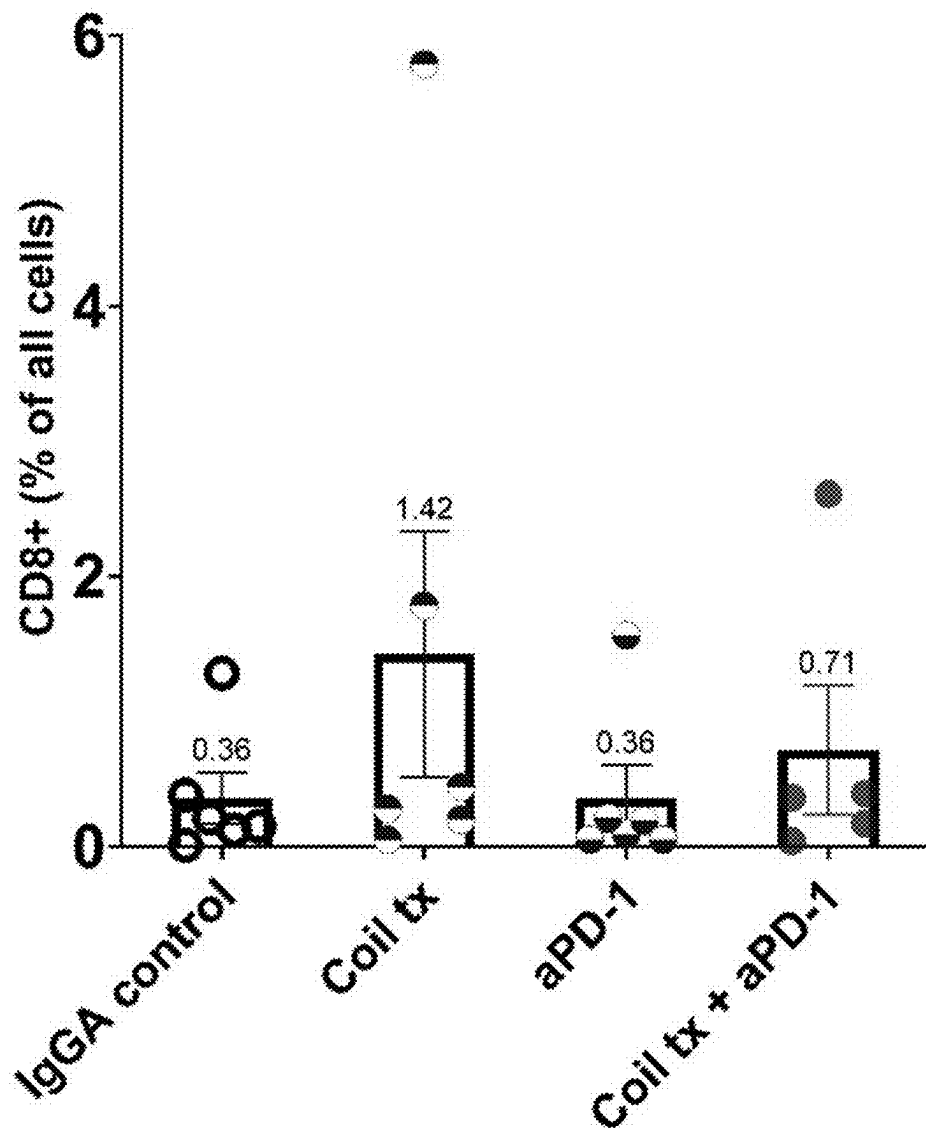


FIGURE 15

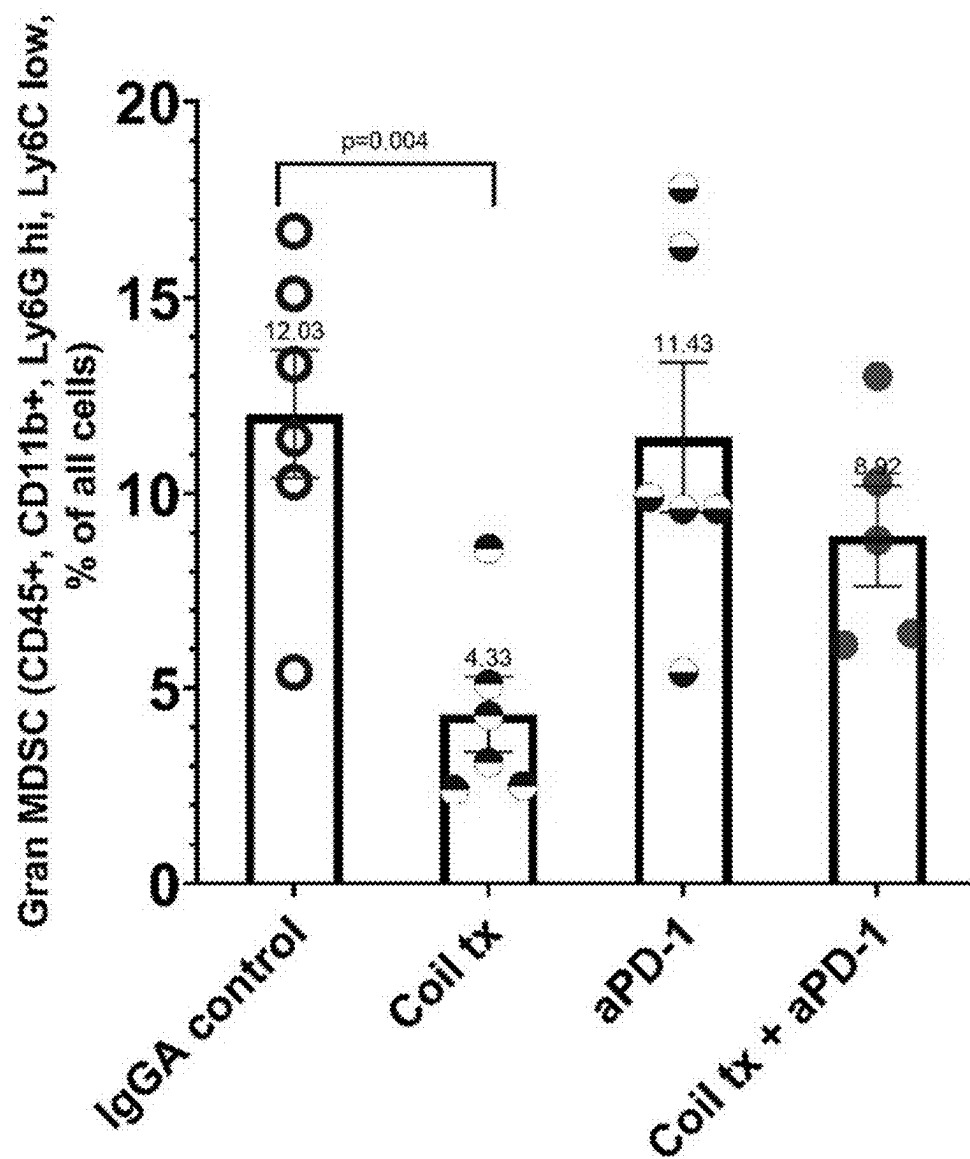


FIGURE 16

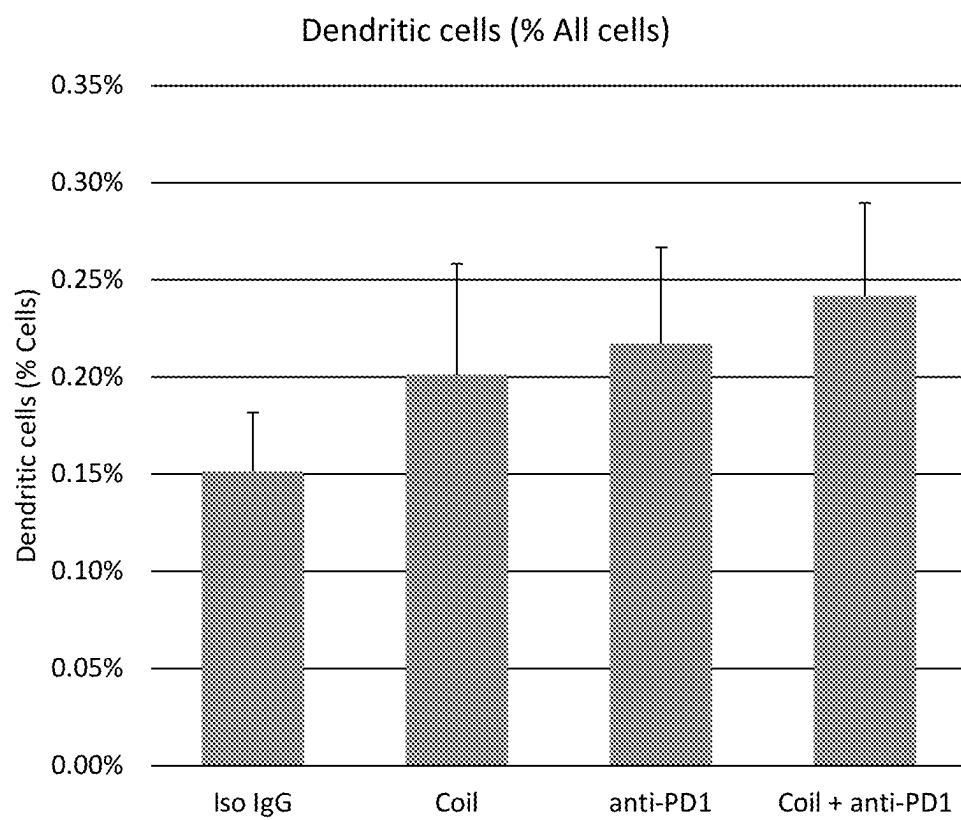


FIGURE 17

SYSTEM AND METHODS FOR TREATING CANCER CELLS WITH ALTERNATING POLARITY MAGNETIC FIELDS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 18/171,357, filed Feb. 19, 2023, now U.S. Pat. No. 11,944,837, entitled "System and Methods for Treating Cancer Cells With Alternating Polarity Magnetic Fields," which is a Continuation of U.S. patent application Ser. No. 17/828,007, filed May 30, 2022, now U.S. Pat. No. 11,583,692, which is a Continuation of U.S. patent application Ser. No. 17/396,608, filed Aug. 6, 2021, now U.S. Pat. No. 11,344,740, which is a Continuation-In-Part of U.S. patent application Ser. No. 17/308,019, filed May 4, 2021, now U.S. Pat. No. 11,344,739, which is a Continuation of U.S. patent application Ser. No. 16/784,239, filed Feb. 6, 2020, now U.S. Pat. No. 11,027,143, which claims the benefit of priority to U.S. Provisional Application No. 62/802,685, filed Feb. 7, 2019. U.S. patent application Ser. No. 17/396,608, filed Aug. 6, 2021 also claims priority to U.S. Provisional Application Ser. No. 63/127,129, filed Dec. 17, 2020, U.S. Provisional Application No. 63/063,198, filed Aug. 7, 2020, and U.S. Provisional Application Ser. No. 62/802,685 filed Feb. 7, 2019. This application claims priority to each of the foregoing applications, each of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention involves treating rapidly proliferating or dividing cells, such as cancer cells, and more specifically to systems and methods for selectively inhibiting or destroying rapidly dividing cells by applying an alternating magnetic field having defined characteristics to a target area of a patient's body. Some embodiments of the invention provide a wearable system capable of providing an ambulatory therapy to a non-stationary patient by applying a magnetic field to inhibit or destroy rapidly dividing cells to the target body area.

BACKGROUND OF THE INVENTION

Cell division is a reproductive process in all living systems, including without limitation simple one-celled organisms such as bacteria and protozoa, as well as more complex organisms such as algae, plants, and animals, including humans. The cell division cycle involves a series of events within the cell that leads to a duplication of the DNA of the cell, with one of the duplicate DNA sequences going to each of two daughter cells. Prokaryotic cells are one-celled organisms that lack an enclosed nucleus and reproduce by a cell division process known as fission. More complex organisms with enclosed nuclei are called eukaryotes, whose cells asexually reproduce by a three-part cell division process involving periods known as interphase, mitosis, and cytokinesis. In the reproduction of sexual cells (i.e., egg and sperm) of more complex organisms, mitosis is replaced by meiosis.

During interphase, the parent cell produces nutrients and other components necessary for mitosis, and the DNA is duplicated as loosely packed chromatin. Mitosis involves separation of the duplicated DNA in the nucleus of the eukaryotic cell into two nuclei, each having a complete copy

of the duplicated DNA. In cytokinesis, the cytoplasm, organelles & cell membrane are divided, forming two daughter cells of roughly equal size.

The process of mitosis is further divided into the stages of prophase, prometaphase, metaphase, anaphase, and telophase. In prophase, the DNA duplicated during interphase condenses into discrete long, thin chromosomes having two chromatids joined by a centromere. Each cell has two centrioles, which move to opposite poles of the cell during prophase. Microtubules radiate from near the two centrioles toward the center of the cell, including some which extend to the chromatids and help to separate the two chromatids into separate daughter chromatids. In metaphase, the chromosomes move toward the cell equator and align in the metaphase plane (or equatorial plane). The daughter chromatids separate from each other at the equator during early anaphase by moving along the microtubule spindle fibers toward the centromeres at opposite poles of the cell, a process which elongates the cell. In late anaphase the daughter chromosomes each reach their opposite poles of the cell, and the cell membrane begins to pinch to form the two daughter cells, which is part of cytokinesis, or the process by which the daughter cells are separated. During telophase, the microtubules continue to lengthen and a new nuclear envelope forms around each of the separated daughter chromosomes, each of which has an identical set of chromosomes, and cytokinesis proceeds with further pinching of the two daughter cells toward becoming separate entities. By the end of telophase, the microtubule spindles disappear. Finally, the daughter cells fully separate, completing cytokinesis.

Cancer cells and some non-cancerous cells (e.g., non-malignant tumors) proliferate or grow in an uncontrolled manner in contrast to normal cells. In addition to the extra space such tumors or cells occupy, they may also damage nearby normal cells. Cancer cells may also metastasize, traveling to other locations in the body, where they continue to hyperproliferate and may form new tumors. The rapid growth of tumors and cancer cells results from their rapid rate of cell division compared to normal cells.

Many effective anti-cancer and anti-tumor therapies are based on the fact that cells in the process of dividing are more sensitive to radiation and many drugs than non-dividing cells. Because tumor cells divide much more frequently than normal cells, it is possible, by using therapies that act on tumor cells while they are dividing, to selectively damage or destroy them while leaving normal cells—which divide less frequently—less affected. However, because many types of cancer cells are only slightly more susceptible to radiation and/or chemotherapy agents than normal cells, it is not always possible to selectively affect tumor cells while leaving normal cells unaffected. Consequently, many radiation and chemotherapy agents significantly damage normal cells as well as tumor cells, leading to a significant patient burden (e.g., pain, scarring, organ damage, blood damage, impaired immune system function, etc.) for even "successful" treatments.

In addition to radiation and chemotherapeutic agents, other therapies involving different modes of action have been used to treat tumor cells, including without limitation ultrasonic and electrical therapies. Electrical currents and electrical fields have been used for decades for medical purposes.

One type of electrical therapy involves applying an electrical current through body tissue separated by two or more conductive electrodes. This type of therapy may be used, for example to stimulate or excite muscle or nerve tissue (e.g.,

pacemakers, defibrillators, neurostimulators) or to generate heat within a desired body tissue (e.g., thermal therapies to remodel collagen or to ablate tissue). Electrical therapies involving conductive electrodes may involve direct current or alternating current at a wide range of frequencies (e.g., less than 1 Hz to above 1 MHz). The energy from electrical currents is delivered to tissue based on the electrical conductive characteristics (e.g., resistance, capacitance) of the tissue. Since these properties are similar for both tumor and normal cells, such therapies affect both tumor and normal cells (e.g., destroying both by heat if they are within the current path) in the same manner. At lower frequencies (typically below 20 kHz), the use of conductive electrodes may be used to stimulate muscle or nerve tissue to activate muscle or nerve fibers. At frequencies used in many electrical therapies (e.g., tens of kHz to MHz), stimulation with conductive electrodes is too rapid for stimulation signals to propagate through such tissue and the signals are "shorted."

Another medical use of electrical energy involves the use of insulated electrodes to deliver high frequency electrical energy radiatively or inductively to target tissue. For example, radio frequency (RF) or microwave energy may be applied radiatively to tissue through the air or another electrically insulating material separating the electrodes from the tissue being treated. The effect of this type of electrical energy on living tissue is based on the dielectric properties of the tissue rather than their conductive characteristics.

More recently, insulated electrodes have been used to treat cancer cells and other rapidly proliferating cells by applying AC electric fields at frequencies of 50-500 KHz and electric field strengths of about 10-1000 V/m to a target body area that includes such cells. Such therapy is often referred to as TC ("tumor curing") field or TTF ("tumor treatment field") therapy. In U.S. Pat. No. 6,868,289, which is hereby incorporated by reference in its entirety, a method and apparatus are disclosed for destroying rapidly proliferating cells using insulated electrodes to generate an electric field. At electric field frequencies of 50-500 kHz, the cell membranes of the dividing cells act to concentrate the electric field lines at the cleavage furrow separating the two daughter cells of the dividing cell. The high-density field at the cleavage furrow causes polarized or charged intracellular components within the cell to move toward the high-density field lines at the cleavage furrow, eventually disrupting the cell membrane at the cleavage furrow, and destroying the dividing daughter cells.

In U.S. Pat. No. 8,019,414, which is hereby incorporated by reference in its entirety, a method of killing or destroying cancer cells is disclosed that involves applying an electric field together with another cancer therapy such as radiation or chemotherapy drugs. The electrical field may be a field such as that disclosed in the '289 patent.

The use of electric fields to destroy cancer cells, while effective at certain frequencies and electrical field strengths, is limited in many practical respects. To provide a safe and consistent electrical field strength, the electrodes of systems such as those disclosed in the '298 and '414 patents must be in intimate contact with the tissue (e.g., skin) of the patient at all times during the treatment. To ensure good contact with the patient's skin, it may be necessary to shave all hair from the skin to which the electrodes are coupled. Because the therapy may be delivered for an extended period of time, the electrodes frequently cause skin irritation at the electrode contact site. For example, in one recent study of TTF therapy, forty-three percent (43%) of patients experienced some skin irritation, with 1% reporting severe skin irritation.

The relatively high incidence of skin irritation or pain may prohibit the therapy in sensitive body areas (e.g., breast tissue, etc.). TTF therapy also involves the use of relatively high voltages. For this reason, patients must be careful in performing everyday activities having a risk of water exposure (e.g., showering, exercise (sweating), or even exposure to rain).

The use of electrodes in direct contact with the patient's skin presents a risk of burning or heating of tissue adjacent to the electrodes. Because of this risk (and buildup of dirt, oils, etc.), the electrodes in TTF therapy systems typically require frequent replacement (e.g., twice each week). Patients wearing TTF electrodes on the scalp reported headaches related to wearing the electrodes 24 hours a day.

TTF electrodes must also be placed by trained users (e.g., technicians or physicians). Because the treatment is highly localized (i.e., between the electrodes), precise location of the cancer/tumor must first be performed, and the electrodes must be placed with a high degree of accuracy to create an electric field that passes through it. If the electrodes are slightly off of optimal placement, the treatment may result in suboptimal results.

In addition, although the '289 patent discloses ambulatory embodiments (i.e., embodiments in which the patent can wear and use the system in performing at least some ordinary non-stationary life activities such as walking, driving, shopping, etc.), in practice the power requirements (e.g., high voltages) for generating appropriate electric fields (e.g., at least 10 V/m) result in bulky and/or heavy electronics boxes that must be coupled to the electrodes and thus carried by the patient. One clinical study showed a relatively high rate of falls in patients carrying these cumbersome TTF electronics boxes.

In view of these limitations to TTF systems, there is a need for safer therapies that may be applied for longer durations to destroy cancer or other rapidly-dividing cells. The many problems associated with electrodes also raise a need for new therapies that avoid a risk of skin pain or the need for continuous contact with skin or other tissue. Because the efficacy of the system depends upon how long the electric fields can be applied to the rapidly-dividing cancer cells, less bulky, heavy, and cumbersome systems are needed to permit truly ambulatory, long duration treatments. Finally, there is a need for therapy systems that do not require trained patients or clinicians for setup.

SUMMARY

In one aspect, the present invention provides a method of treating cancer cells in a target body area of a patient, comprising: providing a magnetic field therapy system comprising: an alternating polarity (AP) magnetic field generator; one or more AP electromagnetic coils coupled to the AP magnetic field generator, wherein the one or more AP electromagnetic coils are energized by an electrical signal from the AP magnetic field generator to generate an AP magnetic field having at least a first frequency and a first field strength; and a controller to control at least one of the first frequency and the first field strength of the AP magnetic field generated by the one or more AP electromagnetic coils; coupling the one or more AP electromagnetic coils to the target body area; generating an AP magnetic field having a first frequency of 0.1-500 KHz and a field strength of 0.2-5 mT using the one or more AP electromagnetic coils; applying the generated AP magnetic field to the target body area using the one or more AP electromagnetic coils, wherein the AP magnetic field modifies the tumor microenvironment to

achieve at least one of: increasing the number of CD8+ lymphocytes in the TME; increasing the ratio of CD8+ to total lymphocytes in the TME; increasing the number of CD4+ lymphocytes; increasing the ratio of CD4+ to total lymphocytes in the TME; increasing the number of tumor infiltrating lymphocytes (TILs) in the TME; increasing the number of antigen presenting cells (APCs) in the TME; increasing the concentration of tumor-suppressive cytokines in the TME; decreasing the concentration of tumor-promoting cytokines in the TME; increasing the number of M1 macrophages in the TME; decreasing the number of M2 macrophages in the TME; decreasing one of the number or concentration of myeloid-derived suppressor cells (MDSCs) in the TME; and increasing one of the number or concentration of natural killer (NK) cells in the TME.

In another aspect, the present invention comprises a method of treating cancer cells in a target body area of a patient, comprising: providing a magnetic field therapy system comprising: an alternating polarity (AP) magnetic field generator; one or more AP electromagnetic coils coupled to the AP magnetic field generator, wherein the one or more AP electromagnetic coils are energized by an electrical signal from the AP magnetic field generator to generate an AP magnetic field having at least a first frequency and a first field strength; and a controller to control at least one of the first frequency and the first field strength of the AP magnetic field generated by the one or more AP electromagnetic coils; coupling the one or more AP electromagnetic coils to the target body area; generating an AP magnetic field having a first frequency of 0.1-500 KHz and a field strength of 0.2-5 mT using the one or more AP electromagnetic coils; applying the generated AP magnetic field to the target body area using the one or more AP electromagnetic coils, wherein the AP magnetic field modifies the tumor microenvironment to achieve at least one of: modulating the blood vessels surrounding the cancer cells; modulating the presence of fibroblasts proximate to the cancer cells; modulating immune cell signaling molecules proximate to the cancer cells; modulating the extracellular matrix surrounding the cancer cells; modulating resident host cells; modulating infiltrating host cells; modulating secreted factors proximate to the cancer cells; modulating the proteins surrounding the cancer cells; modulating the presence of pericycles proximate to the cancer cells; and modulating the presence of adipocytes proximate to the cancer cells.

In another aspect, the present invention provides a method of treating cancer cells in a target body area of a patient, comprising: providing at least one electromagnetic coil; providing a controller coupled to the at least one electromagnetic coil; coupling the at least one electromagnetic coil to the target body area; applying to the target body area an alternating polarity (AP) magnetic field having a frequency of 0.5-500 KHz and a field strength of 0.05-5 mT, wherein the AP magnetic field is generated by the at least one electromagnetic coil under the control of the controller, and the AP magnetic field selectively affects the cancer cells to achieve at least one of damaging the cancer cells, inhibiting the growth of the cancer cells, reducing tumor size, inhibiting angiogenesis, eliciting an immune response to the cancer cells, increasing tumor immunogenicity, decreasing immunosuppressive activity of the cancer cells, recruiting one of antigen-presenting cells and immune effector cells to the tumor microenvironment, or preventing metastasis of the cancer cells, while leaving non-cancer cells substantially unharmed.

In another aspect, the present invention provides a system for treating cancer cells in a target body area of a patient comprising: at least one electromagnetic coil coupled to a target body area; and a controller for controlling the at least one electromagnetic coil to generate and apply to the target body area an AP magnetic field having a frequency of 0.1-500 kHz and a field strength of 0.05-5 mT, wherein the AP magnetic field selectively affects the cancer cells to achieve at least one of damaging the cancer cells, inhibiting the growth of the cancer cells, reducing tumor size, inhibiting angiogenesis, eliciting an immune response to the cancer cells, increasing tumor immunogenicity, decreasing immunosuppressive activity of the cancer cells, recruiting one of antigen-presenting cells and immune effector cells to the tumor microenvironment (TME), or preventing metastasis of the cancer cells, while leaving non-cancer cells substantially unharmed.

In another aspect, the present invention provides a system for treating cancer cells in a target body area of a patient comprising: at least one electromagnetic coil coupled to a target body area; a power supply for supplying power to said electromagnetic coil; and a controller for controlling the at least one electromagnetic coil and power supply to generate and apply to the target body area an AP magnetic field having a frequency of 0.1-500 kHz and a field strength of 0.1-5 mT, wherein the AP magnetic field modifies the tumor microenvironment (TME) by at least one of: increasing the number of CD8+ lymphocytes in the TME; increasing the ratio of CD8+ to total lymphocytes in the TME; increasing the number of CD4+ lymphocytes; increasing the ratio of CD4+ to total lymphocytes in the TME; increasing the number of tumor infiltrating lymphocytes (TILs) in the TME; increasing the number of antigen presenting cells (APCs) in the TME; increasing the concentration of tumor-suppressive cytokines in the TME; decreasing the concentration of tumor-promoting cytokines in the TME; increasing the number of M1 macrophages in the TME; decreasing the number of M2 macrophages in the TME; decreasing one of the number or concentration of myeloid-derived suppressor cells (MDSCs) in the TME; and increasing one of the number or concentration of natural killer (NK) cells in the TME.

In another aspect, the present invention provides a system for treating cancer cells in a target body area of a patient comprising: at least one electromagnetic coil coupled to a target body area; a power supply for supplying power to said electromagnetic coil; and a controller for controlling the at least one electromagnetic coil and power supply to generate and apply to the target body area an AP magnetic field having a frequency of 0.1-500 kHz and a field strength of 0.1-5 mT, wherein the AP magnetic field modifies the tumor microenvironment (TME) by at least one of: modulating the blood vessels surrounding the cancer cells; modulating the presence of fibroblasts proximate to the cancer cells; modulating immune cell signaling molecules proximate to the cancer cells; modulating the extracellular matrix surrounding the cancer cells; modulating resident host cells; modulating infiltrating host cells; modulating secreted factors proximate to the cancer cells; modulating the proteins surrounding the cancer cells; modulating the presence of pericycles proximate to the cancer cells; and modulating the presence of adipocytes proximate to the cancer cells.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic block diagram of a system for providing an alternating polarity (AP) magnetic field to a

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target body area of a patient's body, according to one embodiment for selectively destroying cancer cells.

FIG. 2 is a front view of a retaining element comprising a bra having one or more AP electromagnetic coils for providing an AP magnetic field to breast tissue, according to one embodiment of the invention.

FIG. 3 is a front view of a retaining element comprising a hat including one or more AP electromagnetic coils for providing an AP magnetic field to brain tissue, according to one embodiment of the invention.

FIG. 4 is a front view of a retaining element comprising a shirt having one or more AP electromagnetic coils for providing an AP magnetic field to thoracic or abdominal tissue, according to one embodiment of the invention.

FIG. 5 is a front view of a retaining element comprising a neck cuff or collar having one or more AP electromagnetic coils for providing an AP magnetic field to a target area of a patient's body, according to one embodiment of the invention.

FIG. 6 is a front view of a retaining element comprising a bandage having one or more AP electromagnetic coils for providing an AP magnetic field to a target area of a patient's body, according to one embodiment of the invention.

FIG. 7A is a photograph at 10× magnification of untreated B16F10 mouse melanoma cells incubated for 24 hours.

FIG. 7B is a photograph of B16F10 mouse melanoma cells exposed to an AP magnetic field for 24 hours according to one embodiment of the invention.

FIG. 8 is a bar graph showing the reduction in cell counts of B16F10 mouse melanoma cells treated with an AP magnetic field for 24 hours compared to untreated controls.

FIG. 9 is a graph of tumor volume as a function of time from the start of a magnetic field tumor therapy in an experiment according to one embodiment of the invention.

FIG. 10 is a schematic block diagram of one embodiment of a system for providing an alternating polarity (AP) magnetic field to a target body area of a patient's body for treating cancer cells.

FIG. 11 is a table summarizing the results of a number of experiments on cancer cells using alternating polarity (AP) magnetic field tumor (MFT) therapy.

FIG. 12 is a graph showing the increase in tumor volume over time for breast cancer cells in an animal model experiment involving MFT therapy.

FIG. 13 is a graph showing the increase in tumor volume over time for colon cancer cells in animals treated with an immunotherapy alone or combined with MFT therapy.

FIG. 14 is a graph showing relative metastasis of breast cancer cells in animals treated with MFT therapy, an immunotherapy, or combined MFT therapy and immunotherapy.

FIG. 15 is a graph showing relative changes of CD8+ T cells in the tumor microenvironment of breast cancer cells in animals treated with MFT therapy, an immunotherapy, or combined MFT therapy and immunotherapy.

FIG. 16 is a graph showing relative changes of myeloid-derived suppressor cells (MDSCs) in the tumor microenvironment of breast cancer cells in animals treated with MFT therapy, an immunotherapy, or combined MFT therapy and immunotherapy.

FIG. 17 is a graph showing relative changes of dendritic cells (DCs) in the tumor microenvironment of breast cancer cells in animals treated with MFT therapy, an immunotherapy, or combined MFT therapy and immunotherapy.

DESCRIPTION

Exemplary embodiments of the present disclosure are illustrated in the drawings, which are illustrative rather than

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restrictive. No limitation on the scope of the technology or on the claims that follow is to be implied or inferred from the examples shown in the drawings and discussed here.

In some embodiments, the invention provides apparatus and methods for treating a patient having cancer or other rapidly dividing cells (e.g., bacterial infection) in a target body area using alternating polarity (AP) magnetic fields at specified frequencies to destroy or inhibit the proliferation of the rapidly dividing cells. The use of electric fields, including without limitation TTF systems, to treat patients having cancer or other diseases characterized by rapidly-dividing cells has a number of limitations that make treatment for some patients difficult, ineffective, painful, or unsafe. Embodiments of the present invention overcome one or more of these limitations by using AP magnetic fields to treat rapidly-dividing or hyperproliferating cells.

As used herein, the terms "magnetic field tumor (MFT) therapy" and "MFTT" refer to systems and methods for treating cancer or other rapidly-dividing cells with AP magnetic fields at specified frequencies and magnetic field strengths to destroy or inhibit the proliferation of such cells. In various embodiments, the present invention may be used to treat one or more cancers such as throat cancer, thyroid cancer, mouth cancer, nose cancer, salivary gland cancer, lung cancer, lung carcinoid tumors, thymic malignancies, tracheal tumors, pancreatic cancer, liver cancer, stomach cancer, kidney cancer, ovarian cancer, prostate cancer, colon cancer and rectal cancer.

As used herein with respect to one kind of MFT therapy, the term "duty cycle" refers to the fraction of a time period in which an alternating polarity magnetic field is applied to the target body area. The duty cycle may be calculated by the formula

$$\text{duty cycle} = \frac{\text{on time}}{(\text{on time} + \text{off time})}$$

where on time is the period for which the magnetic field is applied to the target body area, and off time is the time following the on time that no magnetic field is applied to the target body area. In one embodiment, the MFT therapy is applied in repeating cycles of on time followed by off time. In a particular example, an 80% duty cycle may involve 8 hours in which an alternating polarity magnetic field is applied to the target body area, followed by 2 hours in which no magnetic field is applied. It will be appreciated that the same duty cycle can be achieved over different time frames, for example 8 minutes on time followed by 2 minutes off time, or 8 days on time followed by 2 days off time.

FIG. 1 is a simplified schematic block diagram illustrating certain components of an MFT therapy system 100 according to an embodiment of the invention. The MFT therapy system 100 includes an alternating polarity magnetic field generator (APMFG) 110 to generate an electrical signal to energize one or more alternating polarity (AP) electromagnetic coils 120 to produce an AP magnetic field having specified frequency and field strength characteristics. In one embodiment, the AP electromagnetic coils 120 may have sizes and shapes adapted to engage one or more target body areas of a patient (e.g., torso, breast, head, neck, throat) for treatment of cancer or hyperproliferating cells in the target body area. The electrical signals generated by APMFG 110 and applied to AP electromagnetic coils 120 is controlled by a controller 130, which specifies the parameters of the magnetic fields to be generated by APMFG 110 and AP

electromagnetic coils **120**, and controls the function and operation of the system **100**. An interface **140** is provided to allow a user to specify treatment parameters to be programmed or communicated to the controller **130**, and to receive information from the controller relating to the operation and status of the MFT therapy system **100**. A power supply **150** provides power to MFT therapy system **100**. Power supply **150** may be selected from a variety of known power supplies, and may comprise, in various embodiments, a battery such as a disposable or rechargeable battery, or a power source such as a standard 120V, 60 Hz electrical power outlet in the US, together with circuitry for regulating the power at appropriate currents and voltages for each of the APMFG **110**, AP electromagnetic coils **120**, controller **130**, and interface **140**.

Controller **130** may include circuitry and other components (e.g., microcontrollers, resistors, registers, memory, firmware, software, etc.) to direct and control the operations of the APMFG **110**, AP electromagnetic coils **120**, and interface **140**. FIG. 1 illustrates an embodiment in which the AP electromagnetic coils **120** are energized directly from the magnetic field generator. In an alternative embodiment (not shown), controller **130** may communicate directly with each of the one or more AP electromagnetic coils **120** to control their operation in whole or in part (e.g., by switches that enable or disable each AP electromagnetic coil **120**).

Controller **130** includes a timing control module **112** for controlling the timing of the MFT therapy delivered by AP electromagnetic coil(s) **120** to one or more target body areas or tissues. In various embodiments, timing control module **112** may cause the AP magnetic field generator **110** and AP electromagnetic coils **120** to provide MFT therapy for a programmed duration such as 1-100 hours or other treatment period, or the timing of and between a plurality of therapy treatment periods. For example, the timing control module **112** may implement a first therapy for a first time period (e.g., during waking hours of the patient) at a first frequency and field strength, followed by a second time period in which no therapy is applied, followed by a third time period in which a second therapy is implemented at a second frequency and second field strength. Timing control module **112** may also control the timing of changes in other treatment parameters, such as changes in the frequency or field strength of the MFT therapy applied to the patient.

A frequency control module **114** controls the frequency of the AP magnetic fields delivered by AP electromagnetic coil(s) **120** to the one or more target body areas or tissues. Frequency control module **114** may control the frequency of the AP magnetic field at a programmed frequency of 0.5-500 kHz. In some embodiments, the frequency control module **114** may control frequency changes to the AP magnetic fields generated by the APMFG **110** and the AP electromagnetic coils **120** at a programmed rate of change or according to specific frequency step changes.

A magnetic field strength control module **116** controls the field strength of the AP magnetic fields applied to the one or more target body areas. Magnetic field strength control module **116** may control the field strength at a programmed magnetic field strength of 0.05-5 mT, and may control changes in the field strength according to a programmed rate of change or programmed step changes in field strength.

Controller **130** may include programming logic, timers, and other circuitry to accomplish the functions of the timing control module **112**, frequency control module **114**, and magnetic field strength control module **116**. It will be appreciated in alternative embodiments, the functions of all or portions of timing control module **112**, frequency control

module **114**, and magnetic field strength control module **116** may be combined into one or more submodules, or implemented by controller **130** as a whole.

In one embodiment, interface **140** may include a user input, such as a keyboard or buttons, to allow a user to input or receive data from controller **130**. In a further embodiment (not shown,) interface **140** may be located within controller **130** and may comprise a transceiver to communicate with a separate user device (not shown) such as a cell phone, tablet, or other computing device to program the MFT therapy system **100** and receive data therefrom (e.g., operating and alarm status flags, programmed parameters, treatment time, etc.). In other alternative embodiments (not shown), interface **140** may be omitted, or may be incorporated as part of a single unit having some or all of the functions of AP magnetic field generator **110**, controller **130**, and interface **140**.

Referring again to FIG. 1, in various embodiments the APMFG **110** may provide an electrical signal to cause each of the one or more AP electromagnetic coils **120** to generate magnetic fields having one or more fixed or variable AP frequencies. Although shown in the simplified schematic diagram of FIG. 1 as coupled to APMFG **110** by a single wire, it will be understood that each of AP electromagnetic coils **120** will generally be coupled to APMFG **110** by a pair of wires (not shown) to provide a complete circuit. In fixed-frequency embodiments, APMFG **110** may cause each of the one or more AP electromagnetic coils **120** to generate a magnetic field having a single frequency or a plurality of frequencies either continuously or intermittently according to a defined duty cycle (e.g., having a programmable on-time during which the magnetic field is emitted from AP electromagnetic coils **120**, followed by an off-time during which no field is emitted). The APMFG **110** may also cause the one or more AP electromagnetic coils **120** to generate AP magnetic fields having a variety of waveforms, e.g., sinusoidal, triangular, trapezoidal etc. In some embodiments the APMFG **110** may cause the one or more AP electromagnetic coils **120** to generate AP magnetic fields having a pre-defined number of waveforms of a specified first frequency, and repeat this pattern at a second specified frequency (burst mode). In other embodiments, the APMFG **110** may cause the one or more AP electromagnetic coils **120** to generate a magnetic field having a waveform which uses a fixed frequency or a combination of frequencies coupled with amplitude modulation.

Whether fixed or variable, the frequency (or frequencies) of the AP magnetic fields generated by each AP electromagnetic coil **120** are preferably frequencies below about 1 MHz, and more preferably are frequencies within a range selected from 0.1-500 KHz, 0.2-400 kHz, 0.5-300 kHz, 1-200 kHz, 5-150 KHz, 10-100 kHz, or 25-100 kHz. In one embodiment, the MFT therapy system **100** may comprise at least two AP electromagnetic coils **120**, each having a fixed or variable frequency within a different frequency range to provide magnetic fields at multiple frequencies to a target body area or tissue. For example, APMFG **110** may generate a first electrical signal to cause a first AP electromagnetic coil **120** to generate an AP magnetic field with a first fixed frequency or a variable first frequency within a first frequency range, and a second electrical signal to cause a second AP electromagnetic coil **120** to generate an AP magnetic field with a second fixed frequency or a variable second frequency within a second frequency range, where both the first frequency range and the second frequency range are ranges within the range of 0.1-500 kHz, 0.2-400 KHz, 0.5-300 kHz, 1-200 kHz, 5-150 kHz, 10-100 KHz, or

25-100 kHz. As a nonlimiting example, the first frequency range may be a low-frequency range (e.g., 1-5 kHz) and the second frequency range may be a higher-frequency range (e.g., 50 kHz-300 kHz).

Without being bound by theory, it is believed that AP magnetic fields within a plurality of frequency sub-ranges within the range of 0.5-500 kHz may affect different aspects of the reproduction cycle of rapidly-dividing cells, and that each such aspect may be more strongly affected by AP magnetic fields within a particular frequency sub-range within the broader range of 0.5-500 kHz. For example, the interruption of angiogenesis by extremely low frequency AP magnetic fields has been reported for AP magnetic fields having a frequency of 50 Hz (Monache et al., "Inhibition of Angiogenesis Mediate by Extremely Low-Frequency Magnetic Fields (ELF-MFs)," PLOS One, 8:11 (November 2013). Different types of cells, including without limitation different types of cancer cells, may require different frequencies for interruption of angiogenesis.

Accordingly, in one embodiment an MFT therapy having a bimodal magnetic field frequency distribution may be applied to the target body area. In one exemplary embodiment, the APMFT 110 may generate a first electrical signal to cause a first AP electromagnetic coil 120 to generate a first variable AP magnetic field distribution that varies the magnetic field frequency over a first time period (e.g., 1 second, 1 minute, 10 minutes, 1 hour) between a first lower limit (e.g., 0.5 kHz) and a first upper limit (e.g., 5 kHz) to broadly interrupt a first metabolic process (e.g., angiogenesis) in a target cell population, as defined by frequency control module 114. The APMFT 110 may also generate a second electrical signal to cause the same or a second AP electromagnetic coil 120 to generate a second variable AP magnetic field distribution that varies the magnetic field frequency over a second time period (e.g., 1 second, 1 minute, 10 minutes, 1 hour) between a second lower limit (e.g., 50 kHz) and a second upper limit (e.g., 400 kHz) to broadly interrupt a second metabolic process (e.g., the mitosis cycle) of rapidly-dividing cells. Additional coils may produce different fixed or variable-frequency AP magnetic fields having different frequencies or frequency ranges to interrupt still other aspects of the reproduction cycle of rapidly-dividing cells. In an alternative example, a single AP electromagnetic coil 120 may be used to sequentially deliver AP magnetic fields within two different AP frequency ranges (e.g., 1-5 kHz for a first treatment period, followed by 50-400 kHz for a second treatment period).

In variable-frequency embodiments, many different ways of implementing a changing frequency are possible, and enumeration herein of specific embodiments of varying frequencies is illustrative and is not intended to be limiting. It will be appreciated that additional variable-frequency embodiments may be implemented in view of the present disclosure. In one embodiment, a magnetic field may be generated having a single frequency that varies from a lower frequency (e.g., 25 kHz) to an upper frequency (e.g., 150 kHz) in a uniform manner (i.e., non-varying rate of frequency change) within a defined frequency range time period or at a desired (e.g., programmed) frequency change rate. In another embodiment, the frequency may vary in a non-uniform manner such as stepwise changes in frequency or different rates of change (e.g., rates of change of frequency are highest near the mid-point between the upper and lower frequency limits). In a still further embodiment, the frequency may vary continuously or intermittently, with variable-frequency periods alternating with non-variable frequency periods. In additional embodiments, a field having

two different frequencies may simultaneously be applied to the target body area (emitted, e.g., by a single coil or by two different coils). By providing multiple (e.g., 2 or more) coils, MFT therapies having a desired frequency distribution (e.g., random, Gaussian, or non-Gaussian) either sequentially or simultaneously may be applied to one or more target areas.

The electrical signal from APMFG 110 to AP electromagnetic coils 120 also defines the field strength of the AP magnetic fields produced by the coils, as defined by magnetic field strength control module 116. MFT therapy systems 100 of the present invention may use relatively low magnetic field strengths to destroy or impair rapidly-proliferating cells. Preferably, MFT therapy fields in systems 100 of the present invention have field strengths of less than 5 milliTesla (i.e., 5,000 μ T), such as field strengths within a range selected from 0.1-5 mT, 0.2-4 mT, 0.4-3 mT, 0.5-2 mT, or 0.8-1.6 mT. In a preferred embodiment, the field strengths are within the range of 0.2-4 mT, and more preferably within the range of 0.5-2 mT. Table 1 summarizes frequency and field strength ranges according to various embodiments of the invention. In various embodiments, additional frequency ranges can be provided using upper or lower boundaries from different ranges provided in the table.

TABLE 1

Frequency range (kHz)	Field strength range (mT)
0.1-500	0.1-5
0.2-400	0.2-4
0.5-300	0.4-3
1-200	0.5-2
5-150	0.1-1.6
10-100	
25-100	

In one embodiment, the magnetic field may have a single, non-varying field strength. Without being bound by theory, it is believed that different cell sizes (e.g., different types of cancers) may require different field strengths for maximum efficacy in destroying or inhibiting cell division. Within such embodiments, however, the AP magnetic fields may have a single, non-varying field strength either continuously or intermittently according to a defined duty cycle as defined by, e.g., timing control module 112 and magnetic field strength control module 116.

In variable-field-strength embodiments, many different ways of varying the field strength can be envisioned, similar to the variations described above respecting frequency changes. As with frequency, enumeration herein of specific embodiments of varying field strength is illustrative, not limiting. Additional variable-field-strength embodiments may be implemented (e.g., by magnetic field strength control module 116) in view of this disclosure. In one embodiment, a magnetic field may have a field strength that varies from a lower limit (e.g., 0.05 mT) to an upper limit (e.g., 1 mT) in a uniform manner (i.e., with a non-varying rate of change) within a defined field strength range time period. In another embodiment, the field strength may vary in a non-uniform manner such as stepwise changes in field strength or with a swept field strength variation with accelerating or decelerating field strength variation (e.g., rates of change of field strength are highest near the upper and lower limits of the field strength range). In a still further embodiment, magnetic field strength may vary continuously or intermittently, with variable-field-strength periods alternating with non-variable-field-strength periods. In some embodiments,

AP magnetic fields at two different frequencies, each having a different field strength, may simultaneously be applied to the target body area (emitted, e.g., by two different AP electromagnetic coils 120). By providing multiple AP electromagnetic coils 120, MFT therapies having a desired frequency and magnetic field strength distribution (e.g., random, Gaussian, or non-Gaussian), either sequentially or simultaneously may be applied to the target body area.

To optimize therapeutic efficacy and patient tolerance, in some embodiments the therapy is suspended for certain periods. This may involve, for example, providing MFT therapy continuously with defined alternating on-time (e.g., a time period within a range of 1 sec-24 hours) and off-time (e.g., 1 sec-24 hours) periods according to a defined treatment duty cycle as defined by timing control module 112. In one embodiment, the on-time and off-time periods may be a time period within a range of 1 second-1 week, 1 sec-24 hours, 1 minute-12 hours, etc.). In one such exemplary embodiment, the MFT therapy is provided continuously at a 10:1 duty cycle by generating and applying the MFT therapy fields for ten (10) minutes, followed by 1 minute in which no therapy is applied, with the process repeated until a predefined total treatment duration (e.g., 2 weeks) is complete. In another embodiment, the same 10:1 duty cycle may be administered by applying the MFT therapy fields for ten hours, followed by a one-hour suspension of therapy, and repeating the process until the total treatment period is complete.

In another embodiment, MFT therapy according to a defined treatment duty cycle comprising on-time and off-time periods may be administered for a defined treatment duration (e.g., 1 minute, 1 hour, 6 hours, 8 hours, 24 hours) after which no further treatment is applied. In a still further embodiment, the MFT therapy may be administered according to the patient's circadian rhythms (e.g., continuously at night or when the patient is sleeping, and according to a defined duty cycle for defined periods during the day such as morning hours, afternoon hours, or evening hours). It will be appreciated that other duty cycles and treatment durations may be used, and that the therapy may involve, as previously discussed, constant or variable magnetic field frequencies and field strengths.

In another embodiment, MFT therapy may be applied according to a defined treatment duty cycle of on-time and off-time periods, with the magnetic field strength varying according to a defined field strength duty cycle. This may involve, for example, a 10:1 treatment duty cycle combined with a 4:1 field strength duty cycle. As a specific example, the MFT therapy may be provided for a 24 hour treatment duration, at a 10:1 treatment duty cycle with AP magnetic fields applied to a target body area for 10 minutes, followed by 1 minute in which no AP magnetic fields are applied. Within the 10-minute treatment periods, 8 minutes may involve variable frequency treatment within a first field strength range of 3.0-4.0 mT, followed by 2 minutes of treatment within a second field strength range of 0.5-1.5 mT, providing a 4:1 field strength duty cycle.

In a further embodiment, the AP magnetic fields in a mode known as "pulse mode" or "burst mode" in which pulses at a defined low frequency (the "pulse frequency") and pulse duration are applied to a target body area, but each pulse comprises a higher frequency magnetic field signal at a frequency within a frequency range and field strength range previously described. In one nonlimiting example, AP magnetic field pulses at a pulse frequency of 50 Hz, with each pulse having a pulse duration of 10 msec, are applied in which each 10 msec pulse comprises pulses having a higher

frequency of 100 kHz. Table 2 summarizes the exemplary pulse frequency and pulse durations for burst mode operation. Frequencies and field strengths within each pulse of Table 2 would be as shown in Table 1.

TABLE 2

Pulse frequency (Hz)	Pulse duration (msec)
0.01-1000	0.1-5000
0.1-500	1-1000
1-200	10-800
2-100	50-500
5-50	60-300
	100-500
	200-1000
	500-2000
	1-10,000
	100-2,000
	1,000-2,000

In alternative embodiments, MFT therapy fields may be applied according to the patient's circadian rhythms, or according to specific times of day. For example, the MFT therapy fields may be applied to the target body area only during daytime hours; only during nighttime hours; during all daytime hours except during mealtime hours; during all daytime hours except when the patient is exercising (as detected by, e.g., an activity monitor); during specific hours of the day (e.g., 9:00 AM-noon and 6:00 PM-5:00 AM). These examples are intended to be exemplary only, and it will readily be appreciated that delivery of MFT therapy fields can be tailored to suspend therapy during certain hours that would be most convenient to the patient, while also minimizing damage to normal (i.e., non-rapidly-dividing) cells.

Most magnetic field-generating coils are constructed so as to generate a magnetic field having an axis along which the magnetic field lines are directed. In some embodiments, multiple AP magnetic coils 120 can be spatially aligned in such a manner that a desired magnetic field distribution is generated in area of interest, such as the entirety or a portion of the target body area.

In some embodiments, one or more parameters defining the MFT therapy (e.g., frequency, field strength, selection of specific coils among a plurality of available coils) may be determined based on the results of certain tests. For example, an imaging procedure may be performed to identify the type and location of the target rapidly-dividing cells (e.g., cancer or tumor cells). In various embodiments, the imaging procedure may be an imaging procedure using one or more of an MRI system, a CT scan system, a PET scan system, and an X-ray system.

Based on the results of the imaging (e.g., cell size(s), type of cells, location of cells, etc.) a healthcare provider such as a physician may select one or more parameters of the MFT therapy, including without limitation, the frequency/frequencies of the magnetic field(s), the field strength(s), the positioning of one or more coils, a coil size, a type of retaining element (e.g., a garment type) to maintain the coils in position relative to the target body area, a duty cycle or schedule for applying therapy, etc. It will be appreciated that the foregoing and other parameters may also be selected based on other tests, e.g., a pathological analysis of the cancer cells such as a microscopic analysis of a biopsy, a chemical test, a genetic test, etc.

In some instances, the results of an imaging procedure prior to the MFT therapy may identify a target body area to which MFTT is to be directed. Based on the location of the

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target body area, in some embodiments a retaining element may be necessary to retain the magnetic coils in a desired position relative to the target treatment area or tissue. Various retaining elements may be used to unobtrusively and securely maintain the magnetic coils in a desired position relative to a desired target area of the patient's body. For example, a bra may be used to house the magnetic coils for treatment of breast cancer cells. In another example, a hat may be used to retain magnetic coils in position to treat brain cancer. In still another example, a neck cuff, collar, or scarf may retain one or more magnetic coils for treatment of esophageal cancer, and a shirt may be used to retain one or more magnetic coils to treat lung cancer. In another embodiment, the retaining element may be a bandage such as an adhesive bandage capable of adhering to the target body area or to skin adjacent thereto. These examples are exemplary and not limiting, and it will be appreciated that a variety of other or additional retaining elements may be used depending upon the target tissue location. Because the magnetic coils do not need to be in direct contact with the skin of the patient, the retaining elements may include pouches or pockets for securely retaining the coils in position with a comfortable and biocompatible lining placed between the coil and the skin or target treatment area. In some embodiments, the AP electromagnetic coils 120 are completely integrated within the retaining element during manufacturing (e.g., the coils are completely integrated inside a garment such as a bra, hat, shirt, bandage, etc.) In various embodiments, the retaining element may include a lead wire for coupling each of the one or more coils 120 to the AP magnetic field generator 110 and/or controller 130. In alternative embodiments, a direct electrical coupling (e.g., a snap fit) may be used between an electronics package and the AP electromagnetic coil(s) 120. The electronics package may include one or more of the power supply 150, controller 130, APMFG 110, and interface 140.

FIG. 2 illustrates a bra 200 that acts as a retaining element for one or more magnetic coils 220 for applying one or more magnetic fields to a target body area to treat cancer cells or other rapidly-dividing cells in breast tissue. Magnetic coils 220 may be the same as coils 120 described in FIG. 1, but may be adapted for placement in bra 200 (e.g., with a size, geometry, etc., for treatment of breast tissue). Bra 200 may in many aspects be constructed similarly to existing bras available at retail clothing outlets, and may include cups 210 for holding breast tissue and retaining coils 220 in position relative to a target body area comprising breast tissue. Straps 230 may be provided to secure the bra 200 to the shoulders of the patient, and side straps or bands 240 for securing the bra to the patient's torso. AP electromagnetic coils 220 may be integrated into bra 200, or may be removably coupled thereto.

One or more cables or wires 250 may be provided to couple each of the coils 220 to an electronics box 260, which may house the remaining components of the MFT therapy system 100 of FIG. 1 such as APMFG 110, controller 130, power supply 150, and in some embodiments interface 140. In alternative embodiments, one or more of the APMFG 110, controller 130, power supply 150, or interface 140 may be provided separately from the electronics box 260. For example, interface 140 may comprise a mobile phone app that communicates directly with one or more of APMFG 110, controller 130, power supply 150, etc., as well as receiving and displaying information from one or more of the foregoing system components. The mobile phone app interface may allow the patient or a healthcare provider to program one or more treatment parameters for the MFT

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therapy system 100, and may display information relating to the MFT therapy or system 100 status (e.g., displaying how long the MFT therapy has been applied, whether a magnetic field is currently being applied to the target tissue from each of the coils 220, the frequency and/or field strength of the currently-provided magnetic fields, remaining battery life, etc.).

FIG. 3 illustrates a hat 300 that acts as a retaining element for one or more AP electromagnetic coils 320 for applying one or more magnetic fields to the treatment of cancer or other rapidly-dividing cells in a target body area comprising brain tissue. Magnetic coils 320 are, in one embodiment, similar to AP electromagnetic coils 120 described in FIG. 1, but may be adapted for placement in hat 300. This may include changes in the size, geometry, or other characteristics to enable effective placement in hat 300 for treatment of brain tissue. In various embodiments, methods and systems of the present invention may be used to treat a variety of brain cancers, including without limitation astrocytomas, glioblastoma multiforme, meningioma, and pituitary tumors. Although depicted as a baseball cap, it will be apparent that many other head coverings, hat types and styles may be used for hat 300 (e.g., skullcap, beret, fedora, etc.). In one embodiment, hat 300 may be a skullcap of an appropriate size to fit closely on the head of the patient, and the magnetic coils 320 may have a concave shape adapted for location or placement in the cap, e.g., inside the hat or in a pocket between an inner and outer layer thereof. AP electromagnetic coils 320 may be integrated into hat 300, or may be removably coupled thereto.

One or more cables or wires 350 may be provided to couple each of the AP coils 320 to an electronics box 360, which may house the remaining components of the MFT therapy system 100 of FIG. 1 such as APMFG 110, controller 130, power supply 150, and in some embodiments interface 140. In alternative embodiments, one or more of the APMFG 110, controller 130, power supply 150, or interface 140 may be provided separately from the electronics box 360. For example, as described in connection with FIG. 2, a separate interface 140 may be provided as a mobile phone app that communicates directly with one or more of APMFG 110, controller 130, power supply 150, etc. Such an app-based interface may also provide information on the MFT therapy to the patient or a healthcare provider (e.g., displaying how long the MFT therapy has been applied, whether a magnetic field is currently being applied to the target tissue from each of the coils 320, the frequency and/or field strength of the currently-provided magnetic fields, remaining battery life, etc.).

FIG. 4 illustrates a shirt 400 that acts as a retaining element for one or more AP electromagnetic coils 420 for applying one or more magnetic fields to the treatment of cancer or other rapidly-dividing cells in a target tissue in a patient's thoracic or abdominal region. This may include, without limitation and depending on the placement of the one or more AP electromagnetic coils 420, treatment of lung cancer, liver cancer, pancreatic cancer, or cancers or tumors in other thoracic or abdominal organs or structures. AP electromagnetic coils 420 are, in one embodiment, similar to coils 120 described in FIG. 1, but may be adapted for placement in shirt 400 based on the target tissue. This may include changes in the coil size, geometry, or other characteristics to enable effective placement in shirt 400 for treatment of the particular target tissue. Although depicted as a T-shirt, that many other types and styles of shirt may be used as shirt 400, including long-sleeve or short sleeve shirts; button, zip, or pullover shirts. In addition, it will be

understood that shirt **400** may comprise other garments that may cover the thoracic or abdominal region of a patient, including sweaters, jackets, coats, etc., although in preferred embodiments a shirt that fits tightly to the patient's body is used to better retain the AP electromagnetic coils **420** in a more precise or controlled placement relative to the target tissue. AP electromagnetic coils **420** may be adapted for location or placement on the inside, outside or in a pocket of shirt **400**, and may be integrated into or removably coupled thereto.

One or more cables or wires **450** may be provided to couple each of the coils **420** to an electronics box **460**, which may house the remaining components of the MFT therapy system **100** of FIG. 1 such as APMFG **110**, controller **130**, power supply **150**, and in some embodiments interface **140**. In alternative embodiments, one or more of the APMFG **110**, controller **130**, power supply **150**, or interface **140** may be provided separately from the electronics box **460**. For example, a separate interface **140** may be provided as a mobile phone app that communicates with one or more of APMFG **110**, controller **130**, power supply **150**, etc. Such an app-based interface may also provide information on the treatment therapy to the patient or a healthcare provider (e.g., displaying how long the therapy has been applied, whether a magnetic field is currently being applied to the target tissue from each of the coils **420**, the frequency and/or field strength of the currently-provided magnetic fields, remaining battery life, etc.).

FIG. 5 illustrates a neck cuff or collar **500** that acts as a retaining element for one or more AP electromagnetic coils **520** for applying one or more magnetic fields to the treatment of cancer or other rapidly-dividing cells in a target tissue in a patient's neck area, including without limitation esophageal cancer, laryngeal cancer, etc. AP electromagnetic coils **520** may be similar to coils **120** described in FIG. 1, but may be adapted for placement in neck cuff or collar **500** based on the target treatment area or tissue. This may include changes in the size, geometry, or other characteristics to enable effective placement in neck cuff **500** and for treatment of the particular target tissue. Neck cuff or collar **500** preferably includes a securing and/or adjustment tab **530** (e.g., Velcro) to adjust the cuff or collar to the patient's size and to secure it in a fixed position relative to the patient's neck. In one alternative embodiment, a neck scarf may be used as a retaining element. AP electromagnetic coils **520** may be adapted for location or placement on the inside, outside or in a pocket of neck cuff or collar **500**, and may be integrated into or removably coupled thereto.

One or more cables or wires **550** may be provided to couple each of the coils **520** to an electronics box **560**, which may house the remaining components of the MFT therapy system **100** of FIG. 1 such as APMFG **110**, controller **130**, power supply **150**, and in some embodiments interface **140**. In alternative embodiments, one or more of the APMFG **110**, controller **130**, power supply **150**, or interface **140** may be provided separately from the electronics box **560**. For example, a separate interface may be provided as a mobile phone app that communicates with one or more of APMFG **110**, controller **130**, power supply **150**, etc. Such an app-based interface may also provide information on the treatment therapy to the patient or a healthcare provider (e.g., displaying how long the therapy has been applied, whether a magnetic field is currently being applied to the target tissue from each of the coils **520**, the frequency and/or field strength of the currently-provided magnetic fields, remaining battery life, etc.).

FIG. 6 illustrates a bandage **600** that acts as a retaining element for one or more magnetic coils **620** for applying one or more magnetic fields to the treatment of cancer or other rapidly-dividing cells in a target tissue anywhere on the body. Magnetic coils **620** may be similar to those described in FIG. 1, but may be adapted for placement in bandage **600** based on the target tissue. This may include changes in the size, geometry, or other characteristics to enable effective placement in bandage **600** and for treatment of any of a variety of different target tissues. Magnetic coils **620** may be adapted for location or placement on the inside, outside or in a pocket of bandage **600**, and may be integrated into or removably coupled thereto.

One or more cables or wires **650** may be provided to couple each of the coils **620** to an electronics box **660**, which may house the remaining components of the MFT therapy system **100** of FIG. 1 such as APMFG **110**, controller **130**, power supply **150**, and in some embodiments interface **140**. In alternative embodiments, one or more of the APMFG **110**, controller **130**, power supply **150**, or interface **140** may be provided separately from the electronics box **660**. For example, a separate interface may be provided as a mobile phone app that communicates with one or more of APMFG **110**, controller **130**, power supply **150**, etc. Such an app-based interface may also provide information on the treatment therapy to the patient or a healthcare provider (e.g., displaying how long the therapy has been applied, whether a magnetic field is currently being applied to the target tissue from each of the coils **620**, the frequency and/or field strength of the currently-provided magnetic fields, remaining battery life, etc.).

Certain embodiments of the retaining element may also provide additional features to enable the MFT therapy to be conveniently delivered to the target body area or tissue. In one embodiment, the retaining element may have integrated magnetic coils **120**, APMFG **110**, and controller **130**, either as separate items in the retaining element or as a single unit. A wire (not shown) may be provided to couple the power supply to one or more of the APMFG **110**, coils **120**, controller **130**, and interface **140**. In one embodiment, the power supply **150** provides power to the controller, which includes circuitry (e.g., rectifiers, converters, transformers, etc.) to modify the electrical power received from the power supply to provide electrical power to controller **130**, which in turn distributes power to the APMFG **110**, AP electromagnetic coils **120**, and interface **140**. In this embodiment, a power supply (e.g., a battery) may be located elsewhere in close proximity to the patient (e.g., in a pocket in the patient's trousers or a jacket).

In some embodiments, the MFT therapy may be provided to a patient in combination with one or more other therapies such as an anti-cancer drug, radiation therapy, or TTF therapy (e.g., therapy as described in U.S. Pat. No. 6,868, 289 or U.S. Pat. No. 8,019,414). The MFT therapy system **100** preferably permits the other (i.e., non-MFT) therapy to be provided at a lower dosage than would be administered in the absence of the MFT therapy to the target body area or tissue, or at a reduced frequency than would be administered in the absence of the MFT therapy, or both. In various embodiments, the co-therapy applied with the MFT therapy may be a drug selected from a chemotherapy drug, a hormone receptor drug, targeted therapy drugs, immunotherapy, angiogenesis inhibitor drugs, a checkpoint inhibitor drug, and a HER2 receptor drug. In other embodiments the co-therapy may be a radiation therapy selected from an internal radiation therapy and an external beam radiation therapy. In still other embodiments, the co-therapy applied

with the MFT therapy may be a TTF therapy involving the application of electrical fields to the target tissue. Without being bound by theory, it is anticipated that one or more co-therapies (or adjuvant therapies), when combined with MFT therapy, may achieve superior results than either therapy when administered alone. In various embodiments, the combination therapy may comprise administering MFT therapy with an anti-cancer drug, radiation, or TTF therapy either simultaneously or sequentially.

It may be desirable in some instances to shield non-target body areas from the MFT therapy fields. In such instances an optional magnetic field shield (not shown) may be provided to shield the non-target areas from the effects of the magnetic fields. In some embodiments, highly localized shields may be provided to shield specific structures within the target body area of the patient, such as specific blood vessels or organ structures that are adjacent to the target rapidly-dividing cells.

It is known that electric fields induce magnetic fields and vice versa. However, without being bound by theory, the present invention involving MFT therapy appears to provide a therapy having a different mode of action than prior art TTF and/or drug therapies. In particular, TTF therapies use capacitive electrodes to induce primarily electric fields at relatively high electric field strengths. According to reported literature (e.g., Kirson et al., Cancer Research 2004) TTF therapies begin to inhibit tumor cell growth at a field strength of about 100 V/m at frequencies of 50-250 kHz. In one recent experiment (see Experiment 1 below), MFT therapies showed surprising results with a similar inhibition of tumor cell growth as reported for TTF therapies but at a fraction (e.g., less than 3%) of typical TTF therapy electric field strengths. In addition, the experimental results discussed hereinafter suggest that efficacious frequencies for TTF and MFT therapies are different as well.

EXPERIMENT 1

Mouse melanoma cells (B16F10 cell line, obtained from the University of California-Berkeley) were incubated in Dulbecco's Modified Eagle Medium (DMEM) in 36 middle wells (5.0 mm diameter) of a 96 well plate for 24 hours at 37° C. The cells in each of the 36 treatment wells were then exposed for 24 hours to an alternating magnetic field at a frequency of 150 kHz and a magnetic field strength of approximately 0.8 mT using a Helmholtz coil, maintained at a temperature of 37° C. Control wells were not exposed to the alternating magnetic field and were incubated at 37° C. for the same time period. After 24 hours, the alternating magnetic field was discontinued and histology was performed for cells in each well (both treatment and control). FIGS. 7A and 7B are illustrative of the differences between typical control and treatment wells. Controls exhibited a significantly higher cell count per well as shown by gross comparison. In addition, control cells maintained a typically angular morphological structure as indicated in FIG. 7A. Magnetic field-treated cells showed significantly decreased cell count and in addition demonstrated rounded morphology indicating cell stress, as shown in FIG. 7B.

FIG. 8 provides a bar chart comparison summarizing the results of treatments performed on two different plates with 36 wells in each plate. Normalizing the cell counts of the control wells as 100, the treated cells showed a reduction of approximately 31%. Although treatment by magnetic fields and electrical fields (e.g., TTF therapy) are fundamentally different (e.g., using coils vs electrodes and generating primarily magnetic vs. electric fields), it is possible to

calculate the strength of the induced electric field from the coils used in Experiment 1 using the equation

$$E_{nc} = \frac{r}{2} \frac{dB}{dt}$$

Using equation 1 yields a maximum induced voltage of 2.34 V/m, or less than 3% of the electric field strengths reported as required for inhibitory activity in TTF therapy. Because Experiment 1 indicates that MFT therapy exhibits effects on cancer cells at such a small fraction of the electrical field strength of TTF therapy, it enables therapies having significant advantages over TTF therapies, including without limitation ambulatory therapies that allow patients to continue many ordinary day-to-day activities without interruption, and minimal encumbrance or burden.

Part of the advantage of MFT therapies over TTF therapies stems from the different hardware configurations of the two systems. While TTF therapies use insulated (e.g., ceramic coated) electrodes, the use of coils instead of electrodes in MFT therapy confers a number of benefits. Because MFT therapy coils—in contrast to the insulated electrodes of TTF therapies—do not need to be in direct contact with the body, MFT coils can be separated from target tissue by one or more clothing layers (e.g., a garment or undergarment). By applying magnetic fields through clothing, MFT therapies provide increased patient comfort and a less cumbersome patient experience.

In addition, MFT therapies can be implemented with significantly less risk to the patient than TTF therapies involving electrodes. The use of coils instead of electrodes results in only a de minimis induced electrical current during MFT therapies, and thus the risk of electrical shorting and consequent uncontrolled heating of patient body tissue is negligible.

Furthermore, because MFT therapies involve coils that can be made relatively small and with minimal current flow through the patient's body, systems for MFT therapies can allow long treatment periods to target cancer and other hyperproliferating cells with little inconvenience to the patient. These and other advantages of MFT therapies over TTF therapies will be more fully appreciated by persons of skill in the art in view of the present disclosure.

EXPERIMENT 2

Method: Balb/c female mice were kept in plastic cages with free access to food and water and under standardized light/dark cycle conditions. Mice were inoculated subcutaneously with 5×10^5 4T1 murine breast carcinoma cells and one week later were randomly divided into the control and therapy groups.

Mice in the therapy group were exposed to therapy on the same day as they were randomized, which is designated as do. The length, width, and height of the tumor were measured with a fine caliper twice each week for the study duration. The tumor volume was expressed as $0.5 \times \text{length} \times \text{width} \times \text{height}$ of the tumor. Antitumor activity was evaluated over the study duration, and the tumor growth inhibition ratio (IR) was calculated at the study end date (d_f) using the equation

$$IR\% = 1 -$$

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-continued

$$(\text{mean } RTV \text{ of drug treated group} / \text{mean } RTV \text{ of control group}) * 100,$$

where

$$RTV(\text{relative tumor volume}) = \frac{\text{tumor volume on } df}{\text{tumor volume on } d0}$$

Results: After the model development, 10 mice were divided randomly into control and therapy groups (n=5 for each arm). Mice in the therapy group were continuously exposed to an alternating polarity magnetic field having a frequency of 26 kHz (± 1 kHz) and a magnetic field strength of approximately 1 mT for 32 days. FIG. 9 shows the measured tumor volume over the study duration for the control and therapy groups. The inhibition ratio (IR) in the therapy group was 25.9%. FIG. 9 suggests that the cancer-inhibiting effects of MFT therapy may be measurable as early as 4 days after the start of the therapy, and that the effects of the therapy become more significant over time, as illustrated by the difference in tumor volume at the end of the study (day 31).

EXPERIMENT 3

Method: As in Experiment 2, balb/c female mice kept in plastic cages with free access to food and water under standardized light/dark cycle conditions were subcutaneously inoculated with 5×10^5 4T1 murine breast carcinoma cells, then randomly divided into control and therapy groups one week later. Mice in the therapy treatment groups were exposed to therapy the same day as randomization (do), and tumor volume and inhibition ratio (IR) were calculated as in Experiment 2.

Results: After the model development, the mice were randomly divided into control and 3 therapy groups (n=6 in each of the control and three therapy groups, 24 mice total). Therapy was provided as an alternating polarity magnetic field having a field strength of 0.8-1 mT (all therapy groups) and frequencies of 26 kHz (± 1 kHz), 51 kHz (± 1 kHz) and 97 kHz (± 1 kHz), respectively, for therapy groups 1, 2 and 3. Therapy was provided continuously for 28 days, at which time the animals were sacrificed. The tumor was measured twice every week and the tumor volume was estimated from these measurements. FIG. 12 shows the tumor growth across all 4 groups.

FIG. 12 suggests that, at least for the 4T1 murine breast carcinoma cells under study, there appears to be a frequency-dependent response, with greatest efficacy at approximately 50 kHz. Without being bound by theory, it is believed that therapies for various cancer cell types may be characterized by a frequency distribution that describes an optimal therapy response. It is also hypothesized that an optimum frequency or frequency distribution may be experimentally determinable for different types of cancer cells, and that therapy may be provided as a single (optimum) frequency that provides the maximum inhibition ratio, or as a frequency distribution including the optimum frequency. Alternatively, alternating polarity magnetic fields at multiple individual frequencies, or multiple overlapping or non-overlapping frequency distributions varying in a random, non-random, or time-dependent distribution (e.g., a Gaussian distribution) may be used to treat cancers of various types.

The MFT therapies of Experiments 2 and 3 suggest that MFTT does not induce toxicity and is tolerable to the patient. A weight loss during a therapy study is often considered as a sign of therapy toxicity. Mice weight was

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typically measured twice a week throughout Experiments 2 and 3, and there was no significant difference in mice weight between the therapy and control arms at therapy end (d_f).

Based on the results of Experiments 2 and 3, in one embodiment the invention comprises a method for treating cancer cells by applying to a target body area or body region an alternating polarity magnetic field having a field strength of from 0.05-5 mT and a frequency within the range of 1 kHz-100 kHz, where the magnetic field selectively affects the cancer cells to achieve at least one of damaging the cancer cells, inhibiting their growth, reducing tumor size, inhibiting angiogenesis, eliciting an immune response to the cancer cells, increasing tumor immunogenicity, decreasing immunosuppressive activity of the cancer cells, recruiting one of antigen-presenting cells and immune effector cells to the tumor microenvironment, or preventing metastasis of the cancer cells. In one embodiment, the therapy leaves non-cancer cells substantially unharmed. In various embodiments, the method may comprise applying to cancer cells in a target body area or region an alternating polarity magnetic field having a frequency within a frequency range selected from 0.2-400 kHz, 0.5-300 kHz, 1-200 kHz, 5-150 kHz, 10-100 kHz, and 25-100 KHz.

In one embodiment, the AP magnetic fields may be applied to the body of the patient using one or more coils, the operation of which may be controlled by a controller that controls the parameters of the magnetic field therapy, which may include frequency, field strength, duty cycle, etc. The controller may also control the operation of various sensors such as body parameter sensors (e.g., temperature sensors, neural activity sensors, etc.) and sensors adapted to detect one or more effects of the MFT therapy, such as ultrasound sensors to detect tumor size, e.g., ultrasound waves delivered to the patient using one or more ultrasound generating elements such as a piezoelectric ultrasound element.

FIG. 10 illustrates a system according to one embodiment of the invention using ultrasound to image the cancer cells in the target body area simultaneously or sequentially with the delivery of a magnetic field tumor therapy (MFTT). Certain components of the system are similar to like components in the system of FIG. 1. For example, coils 1020, magnetic field generator 1010, magnetic field controller 103 (including timing control module 1012, frequency control module 1014, and magnetic field strength control module 1016), interface 1040, and power supply 1050 provide similar functions to those of coils 120, magnetic field generator 110, magnetic field controller 130, timing control module 112, frequency control module 114, magnetic field strength control module 116, interface 1040, and power supply 1050 of FIG. 1. For brevity, discussion of those elements of FIG. 10 is simplified herein.

In larger mammalian subjects, including human patients, in one embodiment the invention comprises delivering an MFT therapy system comprising at least one body-worn electromagnetic coil 1120 and one or more ultrasound elements 1060 coupled to the patient (e.g., in an article of clothing and/or an ultrasound sensor retaining element), and periodically imaging the tumor using the ultrasound elements 1060 (which may comprise, e.g., ultrasound delivery elements and ultrasound sensors). In one embodiment, the system may comprise an ultrasound generator 1065 coupled to the ultrasound elements (e.g., piezoelectric probes and sensors to detect reflected ultrasound energy from body tissue) and to a controller 1070 having software to analyze the ultrasound data and/or images and using the images to interrupt, restart, or adjust therapy, thereby providing a closed-loop therapy.

In the embodiment of FIG. 10, ultrasound controller 1070 includes a timing control module 1072 for controlling the timing of ultrasound images (e.g., obtaining images at a frame rate of 1-500 frames per second for a defined time periods and/or timepoints (such as once every 10 minutes, 5 images every 30 minutes, etc.), an ultrasound probe control module 1074 for controlling the application of ultrasound energy to the target body area and the detection of reflected ultrasound energy from the target via ultrasound elements 1060, and an ultrasound image generation and analysis module 1076 for processing the ultrasound energy reflected from the patient's body to produce ultrasound images of the cancer cells (e.g., a tumor).

Although depicted as separate elements in FIG. 10, it will be appreciated by persons of skill in the art in view of the present disclosure that ultrasound controller 1070 and its substructures may be combined with those of magnetic field controller 1030 in a single controller. In an alternative embodiment, separate controller elements may be provided for some or each of the substructures of the ultrasound controller 1070 and magnetic field controller 1030.

The MFT therapy system of FIG. 10 further comprises a power supply 1080 and one or more interfaces 1090 for a user (e.g., a patient and/or caregiver), although it will be appreciated that power supply 1080 may be combined with power supply 1050 into a single power supply (which may comprise one or more of a battery or standard interface to an AC system such as a wall socket). Likewise, interfaces 1090 and 1040 may be combined into a single interface for a particular user (e.g., patient, caregiver) that provides information on both the MFT therapy and the ultrasound portion of the system (e.g., providing ultrasound images for display to the patient or caregiver).

In one embodiment, the system of FIG. 10 may provide a closed-loop therapy in which the magnetic field controller 1030 uses ultrasound data (e.g., ultrasound images from image generation/analysis module 1076) to automatically change a parameter of the MFT therapy (e.g., the magnetic field strength, the magnetic field frequency, duty cycle, on time, off time, etc.). In another embodiment, the magnetic field controller may use the ultrasound data to take additional actions, such as to communicate a message to the patient or a caregiver, to recommend a change in a therapy parameter of the MFTT or a drug therapy, to turn the MFTT on or off, or to turn an adjunctive therapy (e.g., a TTF therapy or a nanoparticle hyperthermia therapy) on or off, etc.

In one embodiment, ultrasound images of cancer cells and/or a tumors in the target body area may be automatically captured (e.g., using ultrasound elements 1060, ultrasound generator 1065, and ultrasound image generation and analysis module 1076) and transmitted (e.g., automatically via a body-worn computing device coupled to a wireless or cloud-based network) to one or both of the patient and a healthcare provider, who may analyze the images and transmit instructions to a body-worn therapy system to change one or more therapy parameters.

In one embodiment, the invention comprises delivering an MFTT using a system comprising at least one body-worn electromagnetic coil and one or more heating or cooling elements to deliver a magnetic field therapy to a target body area or region of the patient simultaneously or sequentially with a hyperthermia or hyperthermia therapy to heat or cool cancer cells (e.g., a tumor). Hyperthermia therapy, in which tumors are heated, has been shown to inhibit tumors. Without being bound by theory, it is hypothesized that providing both MFTT and hyperthermia therapy may result in more

efficacious therapies for treating cancer cells. In one embodiment, one or more ultrasound delivery elements 1060 (e.g., piezoelectric elements, not shown) may be used as heating elements to provide the hyperthermia elements. In a particular embodiment, the ultrasound elements may be used for both imaging and for providing hyperthermia elements. In other embodiments, separate ultrasound elements may be provided for imaging and for hyperthermia therapy.

In a still further embodiment, the invention comprises providing an MFTT to a patient and using nanoparticles to apply a hyperthermia therapy to a target body area or region comprising cancer cells. In a particular embodiment, nanoparticles may be delivered to a tumor, and heated by the MFTT system by applying one or more of an alternating polarity magnetic field, ultrasound energy, or another heat source (not shown) to the nano particles. The alternating polarity magnetic field used to heat the nanoparticles may have a field strength and frequency as previously described, or may have different field strength and frequency parameters than those used to provide the MFT therapy.

Without being bound by theory, it is believed that MFTT operates in at least one mode of action by favorably altering the tumor microenvironment (TME) and can work synergistically with other anti-cancer therapies, including without limitation immunotherapies. Cell phenotyping of the TME in some experiments (described hereinafter) support this hypothesis.

Tumor microenvironment or TME refers to the area (which may more accurately comprise a volume of tissue in the patient's body) surrounding the cancer cells/tumor that encompasses surrounding blood vessels, fibroblasts, immune cells, signaling molecules, the extracellular matrix, resident and infiltrating host cells, secreted factors, proteins, tumor vasculature and lymphatics, pericytes and sometimes adipocytes. The TME is a complex ecosystem where subtle interactions profoundly impact tumor progression by influencing processes that lead to tumor eradication, increased metastasis, or the establishment of dormant micrometastases. The tumor microenvironment can also shape therapeutic responses and resistance, justifying the recent impetus to target components of the tumor microenvironment, which is best exemplified by the success of immunotherapies in the clinic. Because cancer cells may be present in the body outside of a tumor, the more generic term "cancer microenvironment" may also be used to clarify that methods and systems of the present invention may be used to modulate one or more aspects of the environment surrounding cancer cells. However, the terms are interchangeable, and "cancer microenvironment," "CME," "tumor microenvironment," and "TME" as used herein are each intended to refer to the area as described above, regardless of whether the cancer cells are part of a tumor.

Despite the fact that immunotherapies have revolutionized cancer care in the last ten years, most patients nevertheless do not respond to them. Research efforts are currently underway to develop therapies that can favorably alter the TME to produce and improve patient outcomes, either when used as a monotherapy or when used in conjunction with other cancer therapies (including immunotherapies).

Immune cells in the TME are fundamental determinants of the invasive and metastatic activity of the cancer cells, and of the tumor's fate. A dominant anti-tumor immune cell is the CD8+ T lymphocyte, which can recognize tumor cells in an antigen-specific manner and secrete cytotoxic molecules to kill them directly. Studies have shown that cancer patients with high CD8+ levels have significantly better recurrence-free survival and overall survival rates. However,

before they are able to perform their cytotoxic functions to destroy cancer cells, CD8+ T cells must be primed and “educated” by antigen presenting cells (APCs), which are dendritic cells.

In contrast to the anti-tumor effects of the CD8+ T cells, myeloid-derived suppressor cells (MDSCs) in the TME promote cancer cell growth. MDSCs are a heterogeneous population of various types of immature myeloid cells that recent studies have shown to cause immune suppression and cancer progression through a variety of mechanisms. Studies have established that elevated MDSC levels in cancer patients are associated with shorter overall patient survival and poor disease-free survival and/or recurrence-free survival.

While cells such as the CD8+ T cells and MDSCs show a consistent anti-cancer or pro-cancer activity, other immune cells demonstrate plasticity in the TME, and are capable of either tumor-promoting or tumor-inhibiting activity, depending upon other factors in the TME. For example, while some macrophages (M1) primarily produce pro-inflammatory cytokines that potentiate the anti-tumor immune response, others (M2) can promote immunosuppression, again depending upon the TME.

Analyses of the TME in retrospective cohort studies across different tumors has established a clear correlation between the density of infiltrating immune cells and the patient’s clinical outcome. In general, these studies have established that the presence of the main cellular players orchestrating the cytotoxic anti-tumor immune response (e.g., cytotoxic CD8+ T cells, mature activated dendritic cells (DCs), and tumor-infiltrating lymphocytes (TILs) are associated with a good clinical outcome in the vast majority of tumor types. In contrast, high densities of M2 macrophages and MDSCs are associated with poor prognosis.

MFT In Vitro Testing

To test the effects of MFT therapy, in vitro tests were conducted on a variety of cancer cell lines by exposing the cancer cells to magnetic fields at defined parameters. The growth of the cancer cells exposed to MFT was compared to cell growth of non-exposed control cancer cells. The results indicate that MFT therapy was able to disrupt cell growth in multiple cell lines, but also that the outcome depends on exposure frequency and field strength. In particular, exposure above a certain frequency of approximately 500 kHz does not result in a statistically significant reduction in the rate of cancer cell growth.

FIG. 11 summarizes the results of a number of experiments treating the MB231 cell line of triple negative breast cancer (TNBC), a highly aggressive cancer, and for the B16F10 melanoma cell line. TNBC cells do not have estrogen or progesterone receptors and produce little HER2 protein. Consequently, treatment options for these cancers are limited, primarily to chemotherapies. Moreover, though TNBC may respond to chemotherapy initially, recurrences of TNBC after such response is more frequent than in other breast cancers.

Each row in the table of FIG. 11 shows the results of a series of experiments in which a number of wells having the cell line were exposed to MFT therapy at a particular frequency and field strength. In one experiment, 20 wells on 96 well plates of the MB231 TNBC cells were exposed for 4 days to magnetic fields at a frequency of 525 kHz and 78 μ T (0.078 mT). At the end of the test, MFT therapy-exposed cells showed similar cell growth as control cells. In another experiment, 20 wells on 96 well plates of the MB231 TNBC cells were exposed for 4 days to magnetic fields at a frequency of 745 kHz and 36 μ T (0.036 mT). At the end of

the test, MFT therapy-treated cells showed similar cell growth as control cells. Both lines 1 and 2 suggest that MFT at frequencies exceeding approximately 500 kHz might not be effective.

In another experiment, 20 wells in 96 well plates of the MB231 TNBC cells were exposed for 4 days to AP magnetic fields at a frequency of 20 KHz and 275 μ T (0.275 mT). As indicated in Line 3 of Table 11, the MFT-exposed cells showed a statistically significant reduction in cell growth compared to the control group. This corresponded to a 15% reduction of the MFT-exposed cells vs. controls, with a statistically significant p-value of less than 0.005. In another experiment, B16F10 melanoma cells in 36 wells in a 96 well plate were exposed for 1 day to AP magnetic fields at a frequency of 150 kHz at a field strength of 800 μ T (0.8 mT). Line 4 of Table 11 indicates that the MFT-exposed cells showed a statistically significant reduction in cell growth compared to the control group. This corresponded to a 31% reduction of the MFT-exposed cells vs. controls, with a statistically significant p-value of less than 0.0001.

MFT Therapy Preclinical Studies

MFT therapy has shown promising results in initial pre-clinical studies on two different animal models covering breast (4T1) and colon cancers (CT26), including inhibiting tumor growth rate (FIGS. 12 and 13) and reducing tumor metastasis to distant organs (FIG. 14). These studies also indicate that using MFT therapy in conjunction with an immunotherapy (checkpoint inhibitor anti-PD1 drug) further improves therapy efficacy.

FIG. 12 is a graph showing the increase in tumor volume over a 30-day period for breast cancer cells (4T1 cell line) in an animal model involving in which control and MFT therapy-treated animals were tested as described in connection with Experiments 2 and 3 above. Treatment animals were exposed continuously for 30 days to an AP magnetic field at 26 kHz, 50 kHz, or 100 kHz frequencies and a field strength of 1 mT. Control animals were not exposed to magnetic field therapy. As shown in the graph of FIG. 12, animals treated with MFT at frequencies of 26 and 50 kHz showed a diminished growth in tumor volume at sacrifice at 30 days, while animals treated at a frequency of 100 kHz showed an increase in tumor growth relative to control. This study suggests that the effects of MFT are dependent upon frequency.

FIG. 13 is a graph showing the increase in tumor volume over a 30-day period for colon cancer cells (CT6 cell line) in an animal model comparing the combination of MFT therapy (50 kHz, 1 mT) and a checkpoint inhibitor anti-PD1 drug (an immunotherapy drug) to the checkpoint inhibitor drug alone, and to isotype IgG2a and control (no exposure to MFT or any drug) animals. Balb/c mice kept in plastic cages with free access to food and water under standardized light/dark cycle conditions were subcutaneously inoculated with 2×10^6 CT26 cells, then randomly divided into 4 groups one week later. The graph of FIG. 13 shows that animals treated with immunotherapy showed a diminished growth in tumor volume after 30 days compared to control and isotype animals. Significantly, animals treated with the combination of MFT therapy and the immunotherapy drug show even further diminished growth compared to the immunotherapy drug alone, suggesting a synergistic effect of MFT therapy and immunotherapy.

FIG. 14 is a bar graph showing relative metastasis of breast cancer cells (4T1 cell line) in an animal lung model. MFT therapy (50 kHz, 1 mT) and the anti-PDF checkpoint inhibitor drug were each used alone in separate animals, and in a third cohort animals were treated both MFT therapy and

the anti-PDF checkpoint inhibitor drug. In this experiment, balb/c female mice kept in plastic cages with free access to food and water under standardized light/dark cycle conditions were subcutaneously inoculated with 5×10^5 4T1 murine breast carcinoma cells, then randomly divided into 4 groups one week later. The animals were sacrificed 28 days after initiation of therapy and the metastatic lung metastases were visually counted. The first bar in FIG. 14 shows the average number of lung metastasis nodules in the control group. The second bar shows the number of lung metastasis nodules in animals exposed to MFT therapy alone, and shows a significant reduction compared to control animals. The third bar shows that animals treated with the anti-PDF checkpoint inhibitor drug alone show a decrease in metastatic lung nodules relative to control but not as significant as MFT therapy alone. The fourth bar shows that animals treated with both MFT therapy and immunotherapy exhibited a dramatic decrease in metastatic lung nodules relative to control, MFT alone, and immunotherapy alone. FIG. 14 shows that MFT therapy and immunotherapy appear to be strongly synergistic for inhibiting metastasis for at least some cancers.

MFT Therapy and TME Cell Phenotypes

Studies of the effect of MFT on TME cell phenotypes have shown that MFT therapy favorably alters the TME by increasing the number of CD8+ T cells (FIG. 15), and the number of dendritic cells (FIG. 16), while reducing the number of granular MDSCs (FIG. 17).

FIG. 15 is a bar graph showing CD8+ T cells as a percentage of all cells in the TME in a breast cancer cell (4T1 cell line) animal model. Concentrations of CD8 cells were measured using flow cytometry. As stated earlier, the CD8+ T cells exhibit an anti-cancer effect. Treated animals were exposed to either MFT therapy (50 kHz, 1 mT) alone; the checkpoint inhibitor anti-PD1 drug alone; or both MFT therapy and the checkpoint inhibitor anti-PD1 drug. The first bar shows control animals having a very low presence of CD8+ cells. In contrast, animals exposed to MFT therapy (50 kHz, 1 mT) showed a much higher concentration of CD8+ cells, suggesting that MFT therapy can be used to modify the TME so as to inhibit tumor progression. The checkpoint inhibitor anti-PD1 drug did not show a greater concentration of CD8+ cells relative to the control (Bar 3), and the animals treated with both MFT therapy and checkpoint inhibitor anti-PD1 drug showed an intermediate concentration of CD8+ cells relative to MFT-exposed animals and control animals.

FIG. 16 is a bar graph showing the relative concentration of MDSC cells as a percentage of all cells in the TME in the same breast cancer animal model as FIG. 15 (4T1 cell line). MDSCs, as previously noted, correlate with poor patient survival. As in FIG. 15, treated animals were exposed to either MFT therapy (50 kHz, 1 mT) alone; the checkpoint inhibitor anti-PD1 drug alone; or both MFT therapy and the checkpoint inhibitor anti-PD1 drug.

Control animals (Bar 1) showed, as expected, a relatively high concentration of MDSCs, as did animals treated with the checkpoint inhibitor anti-PD1 drug alone (Bar 3). Animals exposed to MFT therapy (50 kHz, 1 mT) showed a significantly diminished relative concentration of MDSCs than control or drug-treated animals. Animals treated with both MFT therapy and the checkpoint inhibitor anti-PD1 drug showed an intermediate concentration of MDSCs, between the MFT therapy animals and the control or drug-treated animals. FIGS. 15 and 16 suggest that MFT shows strong enhancement of certain anti-cancer cells and/or

reduction of pro-cancer cells in the TME, an effect that is somewhat reduced by co-therapy with the drug.

FIG. 17 is a bar graph showing the relative concentration of dendritic cells (DCs) as a percentage of all cells in the TME in the same breast cancer animal model as FIGS. 15 and 16 (4T1 cell line). DCs exhibit an anti-cancer/antitumor effect and correlate with improved patient survival. As in FIGS. 15 and 16, treated animals were exposed to either MFT therapy (50 kHz, 1 mT) alone; the checkpoint inhibitor anti-PD1 drug alone; or both MFT therapy and the checkpoint inhibitor anti-PD1 drug.

Control animals (Bar 1) showed a relatively low concentration of dendritic cells. Animals treated with MFT therapy alone (Bar 2) showed an improvement in the concentration of dendritic cells relative to control animals. Bar 3 shows a small increase in dendritic cell concentration relative to MFT-treated animals in animals treated with immunotherapy alone. Animals treated with both MFT therapy and the checkpoint inhibitor anti-PD1 drug, on the other hand, showed a further increase in the concentration of dendritic cells relative to control, MFT-treated animals, and drug-treated animals.

Results from these preclinical studies have shown that MFT therapy inhibits tumor growth and prevents metastasis. Cell phenotyping of TME using flow cytometry indicates that MFT therapy works by favorably altering the TME, in particular by increasing the CD8+ and dendritic cells and decreasing the granular MDSC cells.

In various embodiments, the present invention relates to the subject matter of the following numbered paragraphs.

101. A method of treating cancer cells in a target body area of a patient, comprising:

using an alternating polarity (AP) magnetic field generator comprising:

an alternating polarity (AP) magnetic field generator; one or more AP electromagnetic coils coupled to the AP magnetic field generator, wherein the one or more AP electromagnetic coils are energized by an electrical signal from the AP magnetic field generator to generate an AP magnetic field having at least a frequency and a field strength; and

a controller to control at least one of the frequency and the field strength of the AP magnetic field generated by the one or more AP electromagnetic coils;

coupling the one or more AP electromagnetic coils to the target body area; generating an AP magnetic field having a frequency of 0.5-500 kHz and a field strength of 0.1-500 KHz and a field strength of 0.2-5 mT using the one or more AP electromagnetic coils; and

applying the generated AP magnetic field to the target body area using the one or more AP electromagnetic coils, wherein the AP magnetic field selectively affects the cancer cells to achieve at least one of

increasing the number of CD8+ lymphocytes in the TME;

increasing the ratio of CD8+ to total lymphocytes in the TME;

increasing the number of CD4+ lymphocytes;

increasing the ratio of CD4+ to total lymphocytes in the TME;

increasing the number of tumor infiltrating lymphocytes (TILs) in the TME;

increasing the number of antigen presenting cells (APCs) in the TME;

increasing the concentration of tumor-suppressive cytokines in the TME;

decreasing the concentration of tumor-promoting cytokines in the TME;

increasing the number of M1 macrophages in the TME; decreasing the number of M2 macrophages in the TME;

decreasing one of the number or concentration of myeloid-derived suppressor cells (MDSCs) in the TME;

increasing one of the number or concentration of natural killer (NK) cells in the TME.

102. The method of numbered paragraph 101, wherein generating an AP magnetic field comprises generating an AP magnetic field having a frequency within a frequency range selected from 0.2-400 kHz, 0.5-300 kHz, 1-200 KHz, 5-150 KHz, 10-100 kHz, and 25-100 kHz; and a field strength within a field strength range selected from of 0.2-5 mT, 0.2-4.0 mT, 0.4-3.0 mT, 0.5-2.0 mT, and 0.1-1.6 mT.

104. The method of numbered paragraph 101, wherein the AP magnetic field selectively affects the cancer cells to achieve at least one of damaging the cancer cells, inhibiting the growth of the cancer cells, reducing tumor size, inhibiting angiogenesis, eliciting an immune response to the cancer cells, increasing tumor immunogenicity, decreasing immunosuppressive activity of the cancer cells, recruiting one of antigen-presenting cells and immune effector cells to the tumor microenvironment, or preventing metastasis of the cancer cells, while leaving non-cancer cells substantially unharmed.

105. The method of numbered paragraph 101, further comprising: shielding at least one non-target body area from exposure to the AP magnetic field during the step of applying the generated AP magnetic field to the target body area.

106. The method of numbered paragraph 101, wherein coupling the one or more AP electromagnetic coils to the target body area comprises using a retaining element to maintain the AP electromagnetic field generator in a desired position proximate to the target body area, and wherein the retaining element is selected from a garment and a bandage.

107. The method of numbered paragraph 106, wherein the cancer cells are brain cancer cells selected from astrocytomas, glioblastoma multiforme, meningioma, and pituitary tumors, and the retaining element is a head covering selected from a hat, a helmet, and a garment head covering.

201. A method of treating cancer cells in a target body area of a patient, comprising:

providing at least one electromagnetic coil;
providing a controller coupled to the at least one electromagnetic coil;

coupling the at least one electromagnetic coil to the target body area;

applying to the target body area an alternating polarity (AP) magnetic field having a frequency of 0.5-500 KHz and a field strength of 0.05-5 mT, wherein the AP magnetic field is generated by the at least one electromagnetic coil under the control of the controller, and the AP magnetic field selectively affects the cancer cells to achieve at least one of damaging the cancer cells, inhibiting the growth of the cancer cells, reducing tumor size, inhibiting angiogenesis, eliciting an immune response to the cancer cells, increasing tumor immunogenicity, decreasing immunosuppressive activity of the cancer cells, recruiting one of antigen-presenting cells and immune effector cells to the tumor microenvironment, or preventing metastasis of the cancer cells, while leaving non-cancer cells substantially unharmed.

202. The method of numbered paragraph 201, wherein applying an AP magnetic field comprises applying an AP magnetic field having a frequency of 1-400 KHz and a field strength of 0.2-2 mT.

203. The method of numbered paragraph 201, wherein applying an AP magnetic field comprises applying an AP magnetic field having a frequency of 10-100 kHz and a field strength of 0.5-1.5 mT.

204. The method of numbered paragraph 201, wherein applying an AP magnetic field comprises applying the AP electrical field to the target body area according to at least one of a treatment duty cycle and a field strength duty cycle.

205. The method of numbered paragraph 204, wherein the treatment duty cycle comprises alternating periods of an on-time in which the AP magnetic field is applied to the target tissue, and an off-time in which the AP magnetic field is not applied to the target tissue.

206. The method of numbered paragraph 204, wherein the field strength duty cycle comprises alternating periods in which the AP magnetic field is applied to the target tissue for a first time period at a first field strength followed by a second time period at a second field strength.

207. The method of numbered paragraph 201, wherein the AP magnetic field comprises a bimodal magnetic field frequency distribution comprising a first variable AP magnetic field that varies the magnetic field frequency between a first lower limit and a first upper limit and a second variable AP magnetic field that varies the magnetic field frequency between a second lower limit and a second upper limit.

208. The method of numbered paragraph 201, wherein the AP magnetic field comprises at least one of a variable frequency and a variable field strength.

209. The method of numbered paragraph 1, further comprising administering to the patient at least one additional anti-cancer therapy selected from an anti-cancer drug, a radiation therapy, and TTF therapy.

210. The method of numbered paragraph 209, wherein administering a TTF therapy comprises applying at least one AC electrical field to the target tissue, wherein the AC electrical field comprises a frequency of 50-500 KHz and an electric field strength of about 10-1000 V/m.

The particular embodiments disclosed above are illustrative only, as the invention may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Embodiments of the present invention disclosed and claimed herein may be made and executed without undue experimentation with the benefit of the present disclosure. While the invention has been described in terms of particular embodiments, it will be apparent to those of skill in the art that variations may be applied to systems and apparatus described herein without departing from the concept, spirit and scope of the invention. Examples are all intended to be non-limiting. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope and spirit of the invention, which are limited only by the scope of the claims.

What is claimed is:

1. A system for treating cancer cells in a body of a patient by modifying a cancer microenvironment (TME) comprising:

an alternating polarity (AP) magnetic field generator;
one or more AP electromagnetic coils coupled to the AP magnetic field generator and adapted to be coupled to a first portion of the patient's body and energized by an electrical signal from the AP magnetic field generator to

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- generate and apply to the first portion of the patient's body an AP magnetic field having at least a first frequency of 5-500 kHz and a field strength of 0.2-5 mT;
- a controller adapted to control the AP magnetic field generator, the one or more AP electromagnetic coils, and at least one of the first frequency and the field strength;
- wherein the AP magnetic field is adapted to modify a cancer microenvironment (TME) surrounding the cancer cells by at least one of:
- increasing a number of M1 macrophages in the TME surrounding the cancer cells;
 - decreasing a number of M2 macrophages in the TME surrounding the cancer cells; and
 - decreasing one of a number or a concentration of myeloid-derived suppressor cells (MDSCs) in the TME surrounding the cancer cells.
2. The system of claim 1, wherein the AP magnetic field is further adapted to modify a cancer microenvironment (TME) surrounding the cancer cells by at least one of:
- increasing a number of CD8+ lymphocytes in the TME surrounding the cancer cells;
 - increasing a ratio of CD8+ to total lymphocytes in the TME surrounding the cancer cells;
 - increasing a number of CD4+ lymphocytes surrounding the cancer cells;
 - increasing a ratio of CD4+ to total lymphocytes in the TME surrounding the cancer cells;
 - increasing a number of tumor infiltrating lymphocytes (TILs) in the TME surrounding the cancer cells;
 - increasing a number of antigen presenting cells (APCs) in the TME surrounding the cancer cells;
 - increasing a concentration of tumor-suppressive cytokines in the TME surrounding the cancer cells;
 - decreasing a concentration of tumor-promoting cytokines in the TME surrounding the cancer cells; and
 - increasing one of a number or a concentration of natural killer (NK) cells in the TME surrounding the cancer cells.
3. The system of claim 1, wherein the AP magnetic field is configured to achieve at least one of damaging the cancer cells, inhibiting growth of the cancer cells, reducing tumor size, reducing metastasis of the cancer cells, and inhibiting angiogenesis for the cancer cells, while leaving non-cancer cells substantially unharmed.
4. The system of claim 1, further comprising:
- a retaining element for retaining the one or more AP electromagnetic coils in a desired position relative to the first portion of the patient's body during the application of the AP magnetic field, and wherein the retaining element is selected from a garment and a bandage.
5. The system of claim 4, wherein the one or more AP electromagnetic coils, the retaining element, and the controller are each wearable by a user, and wherein the system comprises an ambulatory treatment system capable of treating the cancer cells while the patient is non-stationary.
6. The system of claim 5, wherein the one or more AP electromagnetic coils comprises a plurality of AP electromagnetic coils, and the retaining element is selected from a hat, a cap, a bra, a shirt, and a vest, and is adapted to maintain the plurality of AP electromagnetic coils in a desired position relative to the first portion of the patient's body.
7. The system of claim 1, wherein the controller comprises:
- a timing control module to perform at least one of:

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- applying the AP magnetic field continuously to the first portion of the patient's body for a first treatment period;
 - applying the AP magnetic field intermittently for a second treatment period in alternating on time periods in which the AP magnetic field is applied to the first portion of the patient's body, followed by off time periods in which the AP magnetic field is not applied to the first portion of the patient's body;
 - applying the AP magnetic field intermittently for one or more circadian treatment periods based on circadian rhythms of the patient; and
 - applying the AP magnetic field intermittently for one or more third treatment periods at defined times of day.
8. The system of claim 1, wherein the controller comprises:
- a frequency control module to perform at least one of:
 - applying an AP magnetic field having a single frequency of 5-500 kHz to the first portion of the patient's body;
 - applying an AP magnetic field having a frequency of 5-500 kHz that varies in a defined pattern.
9. The system of claim 8 wherein the frequency control module further controls the AP magnetic field generator and the one or more AP electromagnetic coils to apply an AP magnetic field having at least one of:
- a frequency of 5-500 kHz that varies randomly;
 - a frequency that varies in a Gaussian distribution in one or more ranges within the range of 5-500 kHz;
 - a frequency that varies in a non-Gaussian distribution in one or more ranges within the range of 5-500 kHz.
10. The system of claim 1, wherein the controller is adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body an AP magnetic field having a frequency of 5-300 kHz and a field strength of 0.4-3 mT.
11. The system of claim 1, wherein the controller is adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body an AP magnetic field having a frequency of 5-200 kHz and a field strength of 0.5-2 mT.
12. The system of claim 1, wherein the controller is adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body an AP magnetic field having a frequency of 5-150 kHz and a field strength of 0.2-1.6 mT.
13. The system of claim 1, wherein the controller is adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body an AP magnetic field having a frequency of 10.0-100 kHz and a field strength of 0.3-5 mT.
14. The system of claim 1, further comprising one or more drugs selected from a chemotherapy drug, an immunotherapy drug, a hormone therapy drug, a targeted therapy drug, and an angiogenesis drug, wherein the each of the one or more drugs is adapted to be administered to the patient one of before, during, or after coupling the one or more AP electromagnetic coils to the first portion of the patient's body, wherein the controller controls the AP magnetic field generator based at least in part on a dosage of the one or more drugs, and wherein the AP magnetic field is adapted to modify the cancer microenvironment (TME) by increasing an efficacy of the one or more drugs.

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15. The system of claim 1, wherein the controller is adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body an AP magnetic field comprising a series of pulse bursts, wherein each pulse in each pulse burst comprises a predefined number of AP magnetic waveforms having a frequency of 5-500 kHz and a field strength of 0.2-5 mT, and each pulse burst in the series of pulse bursts comprises a series of pulses applied at a pulse frequency of 0.01-1000 Hz and a pulse duration of 0.1-5000 msec.

16. The system of claim 1 wherein the AP magnetic field is adapted to modify the cancer microenvironment (TME) of a brain cancer in a brain of the patient, wherein the brain cancer comprises one of cancerous brain tissue and a cancer metastasized to the brain from another region of the patient's body.

17. A system for treating cancer cells in a first portion of the patient's body by modifying a cancer microenvironment (TME), comprising:

an alternating polarity (AP) magnetic field generator adapted to provide an electrical signal to cause an AP magnetic field to be generated;

one or more AP electromagnetic coils coupled to the AP magnetic field generator and adapted to be coupled to the first portion of the patient's body and energized by the electrical signal from the AP magnetic field generator to generate an AP magnetic field having at least a first frequency of 5-500 kHz and a first field strength of 0.2-5 mT, wherein the AP magnetic field is adapted to modify the cancer microenvironment (TME) by modulating immune cell signaling molecules proximate to the cancer cells; and

a controller adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils to cause the one or more AP electromagnetic coils to generate and apply to the first portion of the patient's body the AP magnetic field.

18. The system of claim 17, wherein the AP magnetic field is further adapted to modify the cancer microenvironment (TME) by at least one of:

modulating blood vessels surrounding the cancer cells;
modulating fibroblasts proximate to the cancer cells;
modulating an extracellular matrix surrounding the cancer cells;

modulating resident host cells;
modulating infiltrating host cells;

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modulating secreted factors proximate to the cancer cells;
modulating proteins surrounding the cancer cells;
modulating a concentration of pericycles proximate to the cancer cells; and
modulating a concentration of adipocytes proximate to the cancer cells.

19. The system of claim 17, wherein the cancer cells are brain cancer cells selected from astrocytomas, glioblastoma multiforme, meningioma, and pituitary tumors, the system further comprising:

a retaining element comprising a head covering selected from a hat, a helmet, and a garment head covering for retaining the one or more AP electromagnetic coils in a desired position relative to the first portion of the patient's body during the application of the AP magnetic field.

20. A method for treating cancer cells in a first portion of the patient's body by modifying a cancer microenvironment (TME) using a system comprising an alternating polarity (AP) magnetic field generator, one or more AP electromagnetic coils coupled to the AP magnetic field generator and adapted to receive electrical signals from the AP magnetic field generator to generate an AP magnetic field, and a controller adapted to control the AP magnetic field generator and the one or more AP electromagnetic coils, the method comprising:

coupling the one or more AP electromagnetic coils to the first portion of the patient's body;

using the controller to cause the alternating polarity (AP) magnetic field generator to provide the electrical signals to the one or more AP electromagnetic coils cause the AP electromagnetic coils to generate an AP magnetic field having a first frequency of 5-500 kHz and a first field strength of 0.2-5 mT, wherein the AP magnetic field is adapted to modify the cancer microenvironment (TME) surrounding the cancer cells by at least one of:

increasing a number of M1 macrophages in the TME surrounding the cancer cells;

decreasing a number of M2 macrophages in the TME surrounding the cancer cells;

decreasing one of a number or a concentration of myeloid-derived suppressor cells (MDSCs) in the TME surrounding the cancer cells; and

modulating immune cell signaling molecules proximate to the cancer cells.

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